

Five OEIS Asymptotic Conjectures for Distinct-Partition Norms

Proofs, all-orders logarithmic expansions,
exact-saddle corrections, and inverse
and extreme-value laws

Research report prepared for Vladimir Reshetnikov
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Merged with an independent manuscript on A022629
(batch 73O1: base source 40, member source 48)
and with two further independent manuscripts
(batch 75: sources 75-01 and 75-06)

Central family

$$A_\alpha(q) = \prod_{k \geq 1} (1 + k^\alpha q^k), \quad a_\alpha(n) = [q^n] A_\alpha(q), \quad \alpha > 0.$$

The cases $\alpha = 1, 2, 3, 4, 5$ are [A022629](#), [A092484](#), [A265840](#), [A265841](#), and [A265842](#). All five consulted entries label their leading logarithmic asymptotic as a conjecture. This report proves those formulas and develops a single refinement mechanism for every fixed positive α .

Scope and verification. The results below have detailed English proofs and accompanying exact and numerical checks. They are not claimed to be Lean-verified or peer-reviewed. The literature comparison is targeted, not an exhaustive priority certification. The complete exponentially improved coefficient transseries is not claimed: the report distinguishes its proved logarithmic block, exact-saddle expansion, and Fourier-resonance scale. Source 48, merged here, states the same scope for $\alpha = 1$ and real $\alpha > 0$: no Lean verification, no worldwide priority, and no convergence of the inverse-logarithmic series.

Abstract

For a partition λ into distinct parts, let $N(\lambda)$ be the product of its parts. We analyze the fixed positive norm moments $a_\alpha(n) = \sum_{\lambda \vdash n}^{\text{distinct}} N(\lambda)^\alpha$. A direct argument proves the logarithmic growth formulas still marked conjectural in five OEIS entries. With $K = \sqrt{2n}$, $r = \log K$, and $c = \pi^2/(6\alpha^2)$, we then prove, to every fixed order,

$$\log a_\alpha(n) = \alpha K \left(r - 1 + \frac{c}{r} + \frac{c}{r^2} + \frac{c - c^2/2}{r^3} + \dots \right).$$

An explicit inverse-function construction generates all coefficients; seven are printed. A separate local-limit argument gives multiplicatively accurate exact-saddle asymptotics and all fixed Edgeworth orders. This separation matters: exponentiating a fixed truncation of the logarithmic expansion does not give a relative asymptotic equivalent. We invert the dominant block using the principal Lambert function and obtain asymptotics for the first index attaining a large value. Under the norm-biased measure, the occupation boundary has a logistic profile, while the largest part has a Gumbel limit and is asymptotic to $(1 + 1/\alpha)\sqrt{2n}$. Finally, we prove that the first Fourier resonances have logarithmic modulus of order $-K/(\alpha^3(\log K)^3)$, explaining a striking finite-size limitation visible in the exact computations. Reproducible scripts, source-status notes, and further research questions accompany the proofs.

This report merges two manuscripts that proved these theorems independently: source 40 (the base text, five OEIS sequences, real $\alpha > 0$) and source 48 (A022629 and the same real family, written $\prod(1 + k^s q^k)$). Every shared theorem is printed once and credited to both, with identical coefficients. Source 48 adds a second, elementary proof of the upper bound, a second derivation of all coefficients by an explicit differential operator, explicit centring of the Gumbel law in $\log \sqrt{2n}$, the radius of convergence of $\log A_\alpha$ at the origin, the logarithm of the largest single weight, the entropy factor $\log(a_\alpha(n)/\max N^\alpha) \sim \pi^2 K/(6\alpha r)$, exact data to $n = 6400$ and its own research questions.

Two further manuscripts of batch 75, sources 75-01 and 75-06, prove the same theorems a third and a fourth time, with identical coefficients; those duplicates are recorded in one table, not reprinted. Their new material forms Sections 16–18: sharp signed cumulant scales, a saddle remainder $O((\alpha Ms)^{-J-1})$ with first correction $\sim -3/(8\alpha Ms)$, localization of the integer threshold to $1 + o(1)$, eventual strict log-concavity and fixed-degree Jensen hyperbolicity (source 75-01); and the convolution powers $\prod(1 + k^\alpha q^k)^\mu$ (A297321, A022630, A022631) with μ -independent coefficients, two further inverse coefficients, and an exact rational check of P_8 (source 75-06).

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1 The problem, its significance, and the source audit

1.1 Five conjectures in one natural family

Let $\mathcal{D}_n$ be the set of partitions of n into distinct positive parts. The empty partition has norm 1. For fixed $\alpha > 0$, define

$$N(\lambda) = \prod_{k \in \lambda} k, \quad a_\alpha(n) = \sum_{\lambda \in \mathcal{D}_n} N(\lambda)^\alpha. \quad (1.1)$$

Choosing whether each part occurs gives the locally uniformly convergent product

$$A_\alpha(q) = \sum_{n \geq 0} a_\alpha(n) q^n = \prod_{k \geq 1} (1 + k^\alpha q^k), \quad |q| < 1. \quad (1.2)$$

For integral α the coefficients are integers. One may interpret k^α as the number of colors available for a part of size k , with at most one part of each *size*, regardless of color. This is not the product $\prod_k (1 + q^k)^{k^\alpha}$, which allows several differently colored copies of the same size.

The OEIS entries consulted on October 1, 2026 have the following status [1, 2, 3, 4, 5].

α	Sequence	Date of the conjectural formula
1	A022629	May 8, 2018
2	A092484	December 27, 2020
3	A265840	December 27, 2020
4	A265841	December 27, 2020
5	A265842	December 27, 2020

All five formulas are attributed there to Václav Kotešovec and have the unified form

$$\log a_\alpha(n) \sim \alpha \sqrt{\frac{n}{2}} (\log(2n) - 2). \quad (1.3)$$

The full array of nonnegative integral moments is [A292189](#); its columns $0, 1, \dots, 5$ are identified in that entry [6]. The zeroth column is the ordinary distinct-partition function, but $\alpha = 0$ is a different asymptotic regime and is excluded from our fixed-positive-parameter theorems.

Source 48, which treats A022629 and the real family, makes two further points [1]. Its use of “currently” refers to the entry as retrieved on October 1, 2026, not to an exhaustive determination of the publication history. And a logarithmic equivalent does not determine a multiplicative equivalent: even the constant -2 inside the parentheses of (1.3) is not specified by the relation $\sim$, since changing it leaves the same equivalence. The additive statement that the corresponding $-\sqrt{2n}$ term is correct is sharper, and Proposition 3.2 proves it. For non-integral α , (1.2) is a positive weighted enumeration, and the product converges normally on compact subsets of $|q| < 1$ (source 48).

These sequences are not arbitrary numerical constructions. They simultaneously encode weighted enumeration, moments of a fundamental multiplicative partition statistic, and a conditioned model of independent binary occupancies. Their common structure makes a family theorem more informative than five separate fits to numerical data.

1.2 What is established here, and what is not a priority claim

The conjectural labels in the consulted entries are verifiable facts; the claim that no proof has ever appeared anywhere would be much stronger and is not made. This report supplies self-contained proofs of the displayed formulas, followed by explicit refinements and probabilistic

consequences. Novelty of the refinements should be assessed against the broader literature before publication.

Schneider and Sills survey partition norms, including extremal norms over distinct partitions and their connection with triangular numbers [7]. Bridges and Craig study norm distributions and moments for *unrestricted* partitions [8]; repeated parts change the analytic problem substantially. Rana, Kaur, and Kumar give recurrences and related norm identities for distinct partitions [9]. None of those results is used as an unproved asymptotic input here.

A particularly relevant recent source is Gikunda’s March 2026 dissertation [10]. Its Chapter 4 uses $\prod_k (1 + u^{\log k} q^k)^{b_k}$ and proves log-norm central limit theorems. For $b_k = 1$ and $u = e^\alpha$ this is our generating family. However, the coefficient estimate in Proposition 4.2.5 and the surrounding analysis use a *shrinking* tilt $\log u = O(t^\delta)$ as the saddle parameter t tends to zero. Our tilt $\alpha > 0$ stays fixed. Thus that estimate does not directly supply the fixed-tilt results below. This is a distinction between asymptotic regimes, not a claim that the underlying family has not previously been studied.

Source 48 places the method in the classical saddle-point and probabilistic tradition of asymptotic enumeration; standard background is in Flajolet and Sedgewick [13]. Granovsky and Stark [14] study a broad product framework and, in their Section 3, the related products $\prod (1 + k^{-q} z^k)$ with $q > 0$. Their main theorem assumes parameters $0 < a_k \leq 1$, so substituting $a_k = k^\alpha > 1$ is not a direct application of that theorem. This observation is not a claim that no other general theorem can cover the present case (source 48).

1.3 Relation to the ProveIt repository

The repository snapshot consulted was 1f1981f682b2878bde51a6ad40c22777f362fc05. Its `Analysis/Transseries/README.md` describes the transseries and inversion volumes, accompanying verification programs, and the separate Lean library for asymptotic scales and power-logarithmic monomials [11]. The present report follows that program by separating three precision levels: a logarithmic asymptotic block, a relative exact-saddle expansion, and flat Fourier contributions. Its Lambert- W inversion is given explicitly rather than left as a formal instruction to “reverse the series.”

No repository theorem is assumed here without proof. In particular, the existence of general transseries infrastructure is not treated as a formal certificate for our analytic remainder estimates. The new material is an independent, reproducible research contribution suitable for integration after review; it does not modify the repository.

Source 48 read the repository’s `Analysis/Transseries` material, which consolidates polynomial-logarithmic asymptotics, Lambert-core inversion, error transport, and staircase inversion [12]; those are the organizing ideas it uses. No repository theorem is imported as an unexamined analytic hypothesis: the needed estimates are proved below. A targeted repository search for A022629 returned no match during its preparation. That search and the literature checks are a useful duplication screen, not a proof of originality (source 48).

[Merge note, batch 73O1.] The repository still contains no other treatment of these sequences; this report is the first. Its methodological relative is the partition-number chapter of the volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion`, which treats $p(n)$ (A000041); nothing is imported from it.

[Merge note, batch 75.] Sources 75-01 and 75-06 (Section 15) inspected the ProveIt commit 29aca108e, which held the archives of sources 40 and 48 unopened and predates their placement. Source 75-01’s statement that a repository search for A022629 found nothing was true of that snapshot only; this report is the repository’s treatment, and both sources are merged into it.

1.4 The two sources and this merge

This report is built from two manuscripts of batch 73O1 of ProveIt’s incoming-reports intake, written independently on October 1, 2026.

- **Source 40** (the base text), *Five OEIS Asymptotic Conjectures for Distinct-Partition Norms*: the family $\prod(1 + k^\alpha q^k)$ for real $\alpha > 0$, the five OEIS sequences $\alpha = 1, \dots, 5$, all fixed orders with seven printed coefficients and an eighth checked numerically, the exact-saddle expansion, Lambert- W inversion, the logistic boundary and Gumbel maximum, the fixed-resonance theorem, and finite-size diagnostics to $n = 5000$. Its section order is kept.
- **Source 48**, *The A022629 Conjecture: All-Orders Logarithmic Asymptotics, Saddle-Point Expansions, Inverse Growth, and the Largest Part of a Weighted Partition*: the same theorems for A022629 and for $\prod(1 + k^s q^k)$, real $s > 0$, with six printed coefficients. Its s is source 40’s α ; with that identification its coefficients $c_j(A)$, $A = \pi^2/(6s^2)$, are exactly source 40’s $P_j(c)$, $c = \pi^2/(6\alpha^2)$, and its inverse coefficients and first Edgeworth correction are identical.

Every result, proof, remark, question and non-claim of both manuscripts is printed. A theorem proved by both is printed once, in source 40’s wording, and credited “sources 40 and 48”. Where source 48’s proof is genuinely different, it follows as a “Second proof (source 48)”. Material only source 48 has is marked (source 48): Propositions 3.4 and 3.5, Lemma 3.6, Corollary 8.1, the operator route of Section 7.3, the explicit forms in Sections 9 and 10.3, the computations of Section 12.5, its questions, its OEIS wording, and Appendix C. Text added in the merge is marked [Merge note, batch 73O1.]. Appendix D records the provenance and every choice the merge made.

[Merge note, batch 75.] Batch 75 added two further independent manuscripts, sources 75-01 and 75-06. Their shared results are recorded once, in Table 2 of Section 15, and their new material is Sections 16–19 (marked (source 75-01) and (source 75-06)). Text added in that merge is marked [Merge note, batch 75.], and dated notes on earlier text [*Batch 75, 2 October 2026.*]. Earlier sections are otherwise unchanged, so their numbering is that of the batch-73O1 report.

1.5 Notation across the two sources

The two manuscripts use the same letters for different objects. This report uses source 40’s notation throughout; source 48’s material is translated as follows.

Notion	This report (source 40)	Source 48
power in the weight generating function, coefficients	α $A_\alpha(q), a_\alpha(n)$	s (renamed) $F_s(q), a_s(n); F(q), a(n)$ for A022629
$\sqrt{2n}, \log \sqrt{2n}$	K, r	x, λ
$\pi^2/(6\alpha^2)$; coefficients	$c; P_j(c)$	$A; c_j(A)$
log of the transition site	$s = -W_{-1}(-t/\alpha)$	L
transition site e^s	M	r
$\log A_\alpha(e^{-t})$	$F(t)$	$S_s(t)$
mean, variance	$-F'(t), B = F''(t)$	$m(t), V(t)$
Bernoulli variables, total	$X_k, S = \sum kX_k$	B_k, X_t
log-expansion series	$\mathcal{H}_\alpha(r)$	$\mathcal{H}_s(\lambda)$
boundary operator and series	$\mathcal{T}, \mathcal{C}_\alpha, \mathcal{B}_\alpha$	$\mathcal{T}, \mathcal{C}_s, \mathcal{B}_s$
truncation order (log, Edgeworth)	R, J	K, M
block length (Lemma 5.1)	h	H
largest single weight	$\Pi_\alpha(n)$	$Q_s(n)$
largest part	L_n	M_n
log threshold; Lambert data	$Y; w, R_0, K_0$	$y; w, \ell, x_0$ (and $Y = y/s$)
explicit Gumbel centring	$\beta_\alpha(r)$	$d_s(\lambda)$
small exponent in proofs	σ, β	α

Readings to avoid.

- Source 48's s is the *power* ($= \alpha$); source 40's s , used here, is the *logarithmic transition coordinate* $-W_{-1}(-t/\alpha)$, which source 48 calls L .
- Source 48's $r = e^L$ is the transition *site* (here M); source 40's r , used here, is $\log \sqrt{2n}$ (source 48's λ).
- Source 48's K is a truncation order or a cutoff index; here K is always $\sqrt{2n}$. Source 48's M is an Edgeworth order; here M is always e^s .
- Source 48's $Q_s(n)$ is the largest single weight (here $\Pi_\alpha(n)$), not source 40's mean-relation series $Q_\alpha(s)$. Source 48's Gumbel centring $d_s(\lambda)$ (here $\beta_\alpha(r)$) is not source 40's boundary series $d_\alpha(s)$.
- B is the variance; $\mathcal{B}_\alpha$ is source 48's mean-boundary series.
- Source 48's Edgeworth sum $\mathcal{E}_M$ is cumulative (all weights below $2M$); source 40's $\mathcal{E}_\ell$ is the single weight 2ℓ . Here $1 + \sum_{\ell < J} \mathcal{E}_\ell$ is source 48's $\mathcal{E}_J$.

[Merge note, batch 75.] Sources 75-01 and 75-06 are translated as follows. The layer count μ , the sharp parameter $\varepsilon_\sharp$, the Gaussian inverse n_* , the inverse coefficients Δ_j and the Jensen and Hermite symbols are new in this report.

Notion	This report	Source 75-01	Source 75-06
power in the weight	α	s (renamed)	r (renamed)
number of layers	μ	—	s (renamed)
generating function, coefficients	$A_\alpha(q)^\mu, a_{\alpha,\mu}(n); a_\alpha(n)$	$F_s(q), a_s(n)$	$F_{r,s}(q), a_{r,s}(n)$
$\sqrt{2n}, \log \sqrt{2n}$	$K, r; K_\mu = \sqrt{2n/\mu},$ r_μ	N, ℓ	$m = \sqrt{2n/s}, L$
$\pi^2/(6\alpha^2)$; coefficients	$c; P_j(c)$	$P/6, P = \pi^2/s^2;$ $c_j(P)$	$u/6, u = \pi^2/r^2;$ $b_j(u)$
log transition coordinate, site	$s, M = e^s$	L, M	v, K
log generating function	$F(t), \mu F(t)$	$\Phi_s(t)$	$\mathcal{L}(t)$
Bernoulli cumulant polynomials	$b_j(p)$	$P_j(p)$	$B_j(p)$
Edgeworth terms	$\mathcal{E}_\ell, \mathcal{E}_\ell^{(\mu)}$	E_r	Q_j
small parameters	$\varepsilon = s/M;$ $\varepsilon_\# = 1/(\alpha Ms)$	$\varepsilon = 1/(sML)$	$\varepsilon = v/K$
truncation orders (log, Edgeworth)	R, J	K, J	M, J
boundary-inverse derivatives	$U_m(s)$	$u_L^{(m)}(0), a = L - 1$	$Y_v^{(m)}(0)$
boundary and mean series	$d_\alpha(s), Q_\alpha(s)$	$h_s(L) - \frac{L}{2} + 1,$ $2B_s(L)$	$c(v), 1 + d(v)$
log threshold; Lambert data	$Y; w, R_0, K_0, n_0$	$\log A; w, z_0, N_0, n_0$	$\log Y; w, L_0, m_0, n_0$
threshold index	$N_\alpha(Y), N_{\alpha,\mu}(Y)$	$\mathcal{N}_s(e^Y)$	$N_{r,s}(e^Y)$
inverse coefficients	$\Delta_j(c)$	coefficients of $\mathcal{N}_s/n_0$	$d_j(u)$
Gaussian inverse	$\Psi_0, n_*(Y)$	$\Psi_0, x_0(A)$	—
Jensen, Hermite, local data	$\mathcal{J}_\alpha^{d,n}, H_d; \delta_n, \tau_n, \gamma_k$	$J_s^{d,n}, H_d; \delta_n, A_n,$ g_k	—
largest part; centring	$L_n; b$	—	$M_n; b_n$
smoothing kernel	$g(v)$	$R(v)$	$h(U)$
divisor sum in (12.2)	$C_\alpha(m)$	$b_s(m)$	$\beta_r(m)$

Readings to avoid (batch 75).

- Source 75-01's s and source 75-06's r are the *power* α ; source 75-06's s is the *number of layers* μ . Neither is this report's $s = -W_{-1}(-t/\alpha)$, which source 75-01 calls L and source 75-06 calls v .
- Source 75-06's K is the transition site M , not $\sqrt{2n}$; its L is $\log \sqrt{2n/\mu}$ (here r_μ). Source 75-01's K is a truncation order and its L is s .
- Source 75-01's $P_j(p)$ are cumulant polynomials (here $b_j(p)$), not the coefficients $P_j(c)$; source 75-06's $b_j(u)$ are the coefficients ($b_j(6c) = P_j(c)$), not the cumulant polynomials $b_j(p)$.
- Source 75-06's $c(v)$ is $d_\alpha(s)$ and its $d(v)$ is $Q_\alpha(s) - 1$; its $d_j(u)$ are the inverse coefficients $\Delta_j(c)$ (at $u = 6c$), unrelated to $d_\alpha(s)$.
- Source 75-01's $\varepsilon = 1/(\alpha Ms)$ is written $\varepsilon_\#$ here; $\varepsilon = s/M$ keeps the meaning of (4.8), and $\varepsilon_\# = \varepsilon/(\alpha s^2)$.
- Source 75-01's $x_0(A)$ is the Gaussian inverse, written $n_*(\log A)$ here; it is not source 48's x_0 ($= K_0$).

- Source 75-01's Hermite H_d is written H_d ; H remains the transition width. Its E_r and source 75-06's Q_j are the Edgeworth terms $\mathcal{E}_\ell$, not $Q_\alpha(s)$.
- Source 75-06's $U(x) = tx - r \log x$ is minus the log odds; it is not the Lagrange coefficient $U_m(s)$.

2 Main results and the hierarchy of scales

Throughout the paper, $\alpha > 0$ is fixed. Constants in O -terms may depend on α and on the indicated truncation order. Put

$$K = \sqrt{2n}, \quad r = \log K, \quad c = \frac{\pi^2}{6\alpha^2}. \quad (2.1)$$

Theorem 2.1 (All-orders logarithmic expansion; sources 40 and 48). *There are explicit polynomials $P_j(c) \in \mathbb{Q}[c]$ such that, for every fixed integer $R \geq 1$,*

$$\log a_\alpha(n) = \alpha K \left(r - 1 + \sum_{j=1}^R \frac{P_j(c)}{r^j} + O_{\alpha,R}(r^{-R-1}) \right). \quad (2.2)$$

The first seven are

$$\begin{aligned} P_1(c) &= c, & P_2(c) &= c, \\ P_3(c) &= c - \frac{1}{2}c^2, & P_4(c) &= c + \frac{9}{10}c^2, \\ P_5(c) &= c + \frac{41}{5}c^2 + \frac{1}{2}c^3, & & \\ P_6(c) &= c + 27c^2 + \frac{1973}{210}c^3, & & \\ P_7(c) &= c + \frac{129}{2}c^2 + \frac{8109}{70}c^3 - \frac{5}{8}c^4. & & \end{aligned} \quad (2.3)$$

Equations (7.9)–(8.3) below generate every P_j .

In particular,

$$\begin{aligned} \log a_\alpha(n) &= \alpha K(r - 1) + \frac{\pi^2 K}{6\alpha r} + \frac{\pi^2 K}{6\alpha r^2} \\ &\quad + \left(\frac{\pi^2}{6\alpha} - \frac{\pi^4}{72\alpha^3} \right) \frac{K}{r^3} + O_\alpha(K/r^4). \end{aligned} \quad (2.4)$$

This proves (1.3) for all five OEIS sequences, and more strongly identifies the $-\alpha K$ term with an error $O_\alpha(K/r)$.

Source 48 proves Theorem 2.1 independently (its Theorem 1.1, with $P_1, \dots, P_6$ written $c_1(A), \dots, c_6(A)$), and adds three remarks. First, its constants are uniform when α ranges over a fixed compact subinterval of $(0, \infty)$; they are *not* claimed to be uniform as $\alpha \downarrow 0$. Second, the case $R = 0$ states $\log a_\alpha(n) = \alpha K(r - 1) + O_\alpha(K/r)$, which already proves more than the displayed OEIS conjecture, and the first nonzero correction is positive:

$$\lim_{n \rightarrow \infty} \frac{r}{K} (\log a_\alpha(n) - \alpha K(r - 1)) = \frac{\pi^2}{6\alpha}. \quad (2.5)$$

Third, all-order statements in this report mean: *for each fixed truncation order, the asserted remainder bound holds*. No convergence of the infinite inverse-logarithmic series is assumed (source 48).

[Merge note, batch 75.] Sources 75-01 (its Theorem 1.1, five coefficients $c_j(P)$, $P = 6c$) and 75-06 (its Theorem 6.1, eight coefficients $b_j(u)$, $u = 6c$) prove Theorem 2.1 independently; $c_j(6c) = b_j(6c) = P_j(c)$ (Table 2). Source 75-06 proves it also for the convolution powers $A_\alpha(q)^\mu$, with the same P_j (Theorem 18.1).

For a different level of precision, define

$$F(t) = \sum_{k \geq 1} \log(1 + k^\alpha e^{-tk}), \quad t > 0, \quad (2.6)$$

and let $t = t_n$ be the unique solution of $-F'(t) = n$. Set

$$s = -W_{-1}(-t/\alpha), \quad M = e^s, \quad t = \alpha s/M, \quad H = \frac{M}{\alpha(s-1)}. \quad (2.7)$$

These definitions use the large real branch $s > 1$, valid for all sufficiently large n . Let $B = F''(t)$ and $\kappa_j = (-1)^j F^{(j)}(t)$.

Theorem 2.2 (Exact-saddle coefficient expansion; sources 40 and 48). *As $n \rightarrow \infty$,*

$$M \sim K, \quad s \sim r, \quad B \sim \frac{M^3}{\alpha s}, \quad a_\alpha(n) \sim \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}}.$$

More precisely, for every fixed $J \geq 1$,

$$a_\alpha(n) = \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}} \left(1 + \sum_{\ell=1}^{J-1} \mathcal{E}_\ell + O_{\alpha,J}((s/M)^J) \right), \quad (2.8)$$

where each $\mathcal{E}_\ell = O_{\alpha,\ell}((s/M)^\ell)$ is given explicitly in (6.2). The first correction is

$$\mathcal{E}_1 = \frac{\kappa_4}{8B^2} - \frac{5\kappa_3^2}{24B^3}. \quad (2.9)$$

Theorem 2.3 (Asymptotic inversion; sources 40 and 48). *For $Y \rightarrow \infty$, let $N_\alpha(Y)$ be the least integer n such that $a_\alpha(n) \geq e^Y$. Define*

$$w = W_0\left(\frac{Y}{\alpha e}\right), \quad R_0 = 1 + w, \quad K_0 = \frac{Y}{\alpha w}. \quad (2.10)$$

Then

$$N_\alpha(Y) = \frac{K_0^2}{2} \left(1 - \frac{2c}{R_0^2} - \frac{2c}{R_0^3} + \frac{4c^2 - 2c}{R_0^4} + O_\alpha(R_0^{-5}) \right). \quad (2.11)$$

A formal recurrence in Section 9 gives all inverse-logarithmic orders, with a remainder at every fixed order.

Theorem 2.4 (Logistic boundary and Gumbel extreme; sources 40 and 48). *Give $\mathcal{D}_n$ the probability law $\mathbb{P}_n(\lambda) = N(\lambda)^\alpha / a_\alpha(n)$. For each fixed $x \in \mathbb{R}$, a part near $M + xH$ has limiting occupation probability $(1 + e^x)^{-1}$; any fixed finite collection of distinct such sites is asymptotically independent. If L_n is the largest part, define*

$$b = b(t_n) > \alpha/t_n, \quad \frac{b^\alpha e^{-t_n b}}{t_n} = 1. \quad (2.12)$$

Then, for every fixed real x ,

$$\mathbb{P}_n\{t_n(L_n - b) \leq x\} \longrightarrow \exp(-e^{-x}), \quad \frac{L_n}{\sqrt{2n}} \longrightarrow 1 + \frac{1}{\alpha} \quad \text{in probability.} \quad (2.13)$$

A precision warning. The error in (2.2) is $O(K/r^{R+1})$, which still grows without bound for fixed R . Consequently, exponentiating a fixed truncation need not approximate $a_\alpha(n)$ to relative error $o(1)$. Formula (2.8) does, because it retains the *exact* saddle and the *exact* infinite sums. There is also a smaller exponential scale from noncentral Fourier arcs; Section 11 identifies it. We do not conflate these three statements into an unjustified single explicit coefficient transseries.

Source 48 draws the same line: the inverse-logarithmic expansion determines the complete block on the scales $Kr, K, K/r, K/r^2, \dots$; it does *not* resolve every smaller block or every exponentially small contribution, and a finite truncation of this block generally leaves a diverging error in $\log a_\alpha(n)$. A multiplicative equivalent is instead obtained from the exact saddle and exact convergent sums. Its numerical routines are independent checks, not substitutes for the proofs; they use arbitrary-precision arithmetic but not directed-rounding interval certification (source 48).

[**Merge note, batch 75.**] Sources 75-01 and 75-06 prove Theorems 2.2, 2.3 and (75-06) 2.4 independently (Table 2). Source 75-01 sharpens the remainder of Theorem 2.2 to $O((\alpha Ms)^{-J})$ (Theorem 16.3) and localizes the threshold of Theorem 2.3 to $1 + o(1)$ (Theorem 16.4); source 75-06 proves Theorems 2.3 and 2.4 for μ layers (Section 18). Both draw the same line between the logarithmic block, the relative expansion and the unresolved exponentially small sectors.

3 A short proof of all five posted conjectures

This section already proves the formulas that are explicitly conjectural in the OEIS entries. Later sections supply the stronger refinements.

[**Merge note, batch 75.**] Sources 75-01 (Proposition 2.3 with Lemma 2.2) and 75-06 (Theorem 2.1) give a third and a fourth proof of Proposition 3.2: the same lower-bound partition, and an upper bound by Lemma 3.1 (75-01) or by the positive-part splitting of source 48's second proof (75-06, for μ layers, Theorem 18.1).

Lemma 3.1 (A useful continuum estimate). *For $s \rightarrow \infty$, $M = e^s$, and $t = \alpha s/M$,*

$$F(t) = \alpha M(s/2 - 1) + O_\alpha(M/s). \quad (3.1)$$

Proof. The function $f_t(x) = \log(1 + x^\alpha e^{-tx})$ is nonnegative and unimodal on $(0, \infty)$, tends to zero at both ends, and has maximum $O_\alpha(s)$. For a nonnegative unimodal function, comparison on unit intervals gives

$$\left| \sum_{k \geq 1} f_t(k) - \int_0^\infty f_t(x) dx \right| \leq 2 \sup_x f_t(x) = O_\alpha(s).$$

With $x = Mu$, the exponent is $\alpha[s(1 - u) + \log u]$. Replacing $\log(1 + e^z)$ by $\max(z, 0)$ changes the integral by $O_\alpha(M/s)$: in a fixed neighborhood of $u = 1$, the derivative of $s(u - 1) - \log u$ is comparable with s , and $\int_{\mathbb{R}} \log(1 + e^{-|v|}) dv < \infty$. Outside that neighborhood, split off $0 < u < e^{-s/2}$; its integral is exponentially small times a polynomial in s . On the remaining region away from 1, the absolute exponent grows linearly in s , except for the negligible small lower crossing. Thus the remaining error is smaller than M/s . The lower crossing changes the following signed baseline by $O(1)$:

$$\alpha M \int_0^1 (s(1 - u) + \log u) du = \alpha M(s/2 - 1).$$

The lattice error $O(s)$ is also $O(M/s)$. □

Proposition 3.2 (sources 40 and 48). *For every fixed $\alpha > 0$,*

$$\log a_\alpha(n) = \alpha\sqrt{2n}(\log \sqrt{2n} - 1) + O_\alpha\left(\frac{\sqrt{n}}{\log n}\right). \quad (3.2)$$

Proof. Let m be maximal with $m(m+1)/2 \leq n$, and write $d = n - m(m+1)/2$, so $0 \leq d \leq m$. The distinct partition

$$1 + 2 + \cdots + (m-1) + (m+d)$$

has norm at least $m!$. Therefore

$$\log a_\alpha(n) \geq \alpha \log(m!) = \alpha K(r-1) + O_\alpha(r),$$

where $m = K + O(1)$ and the usual integral bounds, or Stirling's formula, give $\log(m!) = m \log m - m + O(\log m)$.

For the upper bound, nonnegative coefficients give $a_\alpha(n) \leq \exp(F(t) + nt)$ for every $t > 0$. Choose $t = \alpha r/K$, that is, $M = K$ and $s = r$ in Lemma 3.1. Since $n = K^2/2$,

$$F(t) + nt = \alpha K(r/2 - 1) + \alpha Kr/2 + O_\alpha(K/r) = \alpha K(r-1) + O_\alpha(K/r).$$

The two bounds prove (3.2). □

Remark 3.3. The lower-bound partition need not maximize the norm. An exact maximizer is unnecessary. Also, a bare asymptotic equivalence to $\alpha K(r-1)$ would not by itself certify $-\alpha K$ as the second term; the error in (3.2) does certify it.

3.1 Two structural facts and a second proof (source 48)

Why an expansion at the origin cannot simply be used at the saddle.

Proposition 3.4 (source 48). *The Taylor series of the analytic logarithm of A_α at $q = 0$ has radius of convergence $3^{-\alpha/3}$, although A_α is analytic throughout $|q| < 1$.*

Proof. The zeros of $1 + k^\alpha q^k$ have modulus $k^{-\alpha/k}$. Among positive integers k , the maximum of $\log k/k$ occurs at $k = 3$: the function decreases for $k > e$, and $\log 3/3 > \log 2/2$. Hence the closest zeros of the product have modulus $3^{-\alpha/3}$. The remaining factors converge normally and cannot remove a zero. Inside this circle all factors are nonzero, so the product has a single-valued analytic logarithm with value zero at the origin. At a closest zero its logarithm cannot be analytic. This proves the assertion. □

Consequently,

$$\log A_\alpha(q) = \sum_{k \geq 1} \sum_{j \geq 1} \frac{(-1)^{j+1}}{j} k^{\alpha j} q^{kj}$$

is useful as a *formal* identity, but this expansion is not convergent at the positive saddles approaching $q = 1$. Treating it as a convergent expansion there would lose the transition that controls this problem.

The largest individual weight. Let

$$\Pi_\alpha(n) = \max_{\lambda \in \mathcal{D}_n} N(\lambda)^\alpha.$$

Proposition 3.5 (source 48). *For fixed $\alpha > 0$, with $K = \sqrt{2n}$ and $r = \log K$,*

$$\log \Pi_\alpha(n) = \alpha K(r - 1) + O_\alpha(r). \quad (3.3)$$

Proof. Write $n = m(m+1)/2 + d$ with $0 \leq d \leq m$. The distinct partition $(1, 2, \dots, m-1, m+d)$ has weight $((m-1)!(m+d))^\alpha$. Since $m = K + O(1)$, Stirling's estimate yields the required lower bound, with error $O_\alpha(r)$.

For the upper bound set $t = \alpha r/K$. Any distinct partition λ of n satisfies

$$\alpha \sum_{k \in \lambda} \log k \leq tn + \sum_{k \geq 1} (\alpha \log k - tk)_+, \quad u_+ = \max(u, 0).$$

For sufficiently large K , the positive terms are exactly those with $2 \leq k < K$, apart from a possible zero term at an integer endpoint. Summing $\log k$ by Stirling's formula and k by the arithmetic progression formula gives

$$\sum_{k \geq 1} (\alpha \log k - tk)_+ = \alpha K(r/2 - 1) + O_\alpha(r).$$

Adding $tn = \alpha Kr/2$ proves the upper bound. $\square$

Lemma 3.6 (transition width; source 48). *Let $M = e^s$, $t = \alpha s/M$, and $s \rightarrow \infty$. Then*

$$\sum_{k \geq 1} \exp(-|\alpha \log k - tk|) = O_\alpha(M/s). \quad (3.4)$$

Proof. Put $\psi(u) = \alpha \log u - tu$. On $k \geq M/2$, the derivative of ψ is at most $-t/2$ once $s \geq 4$, and $\psi(M) = 0$. Thus this part of the sum is bounded by a two-sided geometric sum of scale $1/t = O_\alpha(M/s)$.

For $1 \leq k \leq M^{1/4}$, use the bound 1, giving $O(M^{1/4})$. On $[M^{1/4}, M/2]$, concavity of ψ implies that its minimum is at an endpoint. For large s , both endpoint values exceed $c_\alpha s$ for a constant $c_\alpha > 0$. This range therefore contributes $O_\alpha(M^{1-c_\alpha})$. Both estimates are $o_\alpha(M/s)$. Adjusting integer endpoints changes nothing. $\square$

Second proof of Proposition 3.2 (source 48). The lower bound follows from $a_\alpha(n) \geq \Pi_\alpha(n)$ and Proposition 3.5 (the same partition as in the first proof). For the upper bound, positivity gives $a_\alpha(n) \leq e^{tn} A_\alpha(e^{-t})$. Use $t = \alpha r/K$ and the exact identity

$$\log(1 + e^u) = u_+ + \log(1 + e^{-|u|}).$$

The positive-part sum was evaluated in the proof of Proposition 3.5; the second sum is bounded by Lemma 3.6 with $M = K$, $s = r$. We obtain

$$tn + F(t) = \alpha K(r - 1) + O_\alpha(K/r + r),$$

and $r = o(K/r)$. This proves (3.2) without any local limit theorem, continuum integral or formal asymptotic manipulation. $\square$

This argument is a separate, elementary proof of the OEIS conjecture; the remaining sections determine what lies inside its error term (source 48).

4 The exact probabilistic saddle and its scales

For fixed $t > 0$, let X_k be independent Bernoulli random variables with

$$p_k = \mathbb{P}_t(X_k = 1) = \frac{k^\alpha e^{-tk}}{1 + k^\alpha e^{-tk}}, \quad S = \sum_{k \geq 1} k X_k. \quad (4.1)$$

The sum is finite almost surely because $\sum p_k < \infty$. Multiplying the Bernoulli probabilities gives the exact identity

$$\mathbb{P}_t(S = n) = a_\alpha(n) e^{-nt - F(t)}. \quad (4.2)$$

Differentiation under the absolutely convergent sums shows

$$\mathbb{E}_t S = -F'(t) = \sum k p_k, \quad \text{Var}_t S = F''(t) = \sum k^2 p_k (1 - p_k) > 0.$$

The mean is continuous and strictly decreasing from ∞ to 0 as t increases from 0 to ∞ . Hence the saddle equation has a unique solution for every $n > 0$.

The large crossing $k^\alpha e^{-tk} = 1$ occurs at $k = M$ in (2.7). Its logarithmic derivative there is

$$\frac{\alpha}{M} - t = -\frac{\alpha(s-1)}{M} = -H^{-1}.$$

Thus H is the width of the transition from predominantly occupied to predominantly unoccupied sites.

Lemma 4.1 (Transition estimates). *For fixed $\alpha > 0$, as $s \rightarrow \infty$,*

$$\sum_{k \geq 1} k p_k \sim M^2/2, \quad (4.3)$$

$$\sum_{k \geq 1} k^2 p_k (1 - p_k) \sim M^2 H \sim M^3/(\alpha s). \quad (4.4)$$

For each fixed integer $j \geq 2$,

$$\sum_{k \geq 1} k^j p_k (1 - p_k) = O_{\alpha,j}(M^{j+1}/s). \quad (4.5)$$

Moreover, for every fixed nonnegative integer h ,

$$\frac{1}{H^{h+1}} \sum_{k \geq 1} |k - M|^h p_k (1 - p_k) \longrightarrow \int_{\mathbb{R}} |x|^h \frac{e^x}{(1 + e^x)^2} dx. \quad (4.6)$$

Proof. For $k = M + xH$ with x in a fixed bounded interval,

$$\alpha \log k - tk = \alpha \log(1 + xH/M) - \alpha s x H/M = -x + O_{\alpha,x}(s^{-2}).$$

Here and below rounding k to an integer changes the scaled coordinate by $O(H^{-1}) = o(1)$. Thus p_k converges to $(1 + e^x)^{-1}$. Since $H \rightarrow \infty$, sums over any fixed x -interval are Riemann sums. The limiting variance density has integral 1.

We record a tail justification so that this argument is not just a local heuristic. On $k/M \in [1/2, 3/2]$, the derivative of the log odds is negative and comparable to $-1/H$ for large s ; the variance density is therefore bounded by $C_\alpha e^{-c_\alpha |k-M|/H}$. On $k \geq 3M/2$, the odds themselves are exponentially small in s and then in tk . On $k \leq M/2$, use $p_k(1 - p_k) \leq \min(1, k^{-\alpha} e^{tk})$, and split at M/s . The range $M/s < k \leq M/2$ is exponentially small in s after scaling by any of the powers of M above. On $k \leq M/s$, the crude bound is $O_\alpha(k^{-\alpha})$; its contribution to

(4.6), even with $|k - M|^h \leq M^h$, is $o(H^{h+1})$. Indeed the resulting ratio is bounded by a fixed power of s times $M^{-\min(\alpha, 1)}$, with an extra logarithm when $\alpha = 1$. These bounds establish uniform integrability for every fixed moment. They also prove (4.5) and (4.4).

Finally, $p_{\lfloor Mu \rfloor} \rightarrow 1$ for $0 < u < 1$ and tends to zero for $u > 1$. The same tail bounds and a Riemann sum give $M^{-2} \sum k p_k \rightarrow \int_0^1 u \, du = 1/2$. $\square$

Source 48 sets up the same Bernoulli representation (its (4.2) is identical, with the total written X_t and the variance V), and the same transition coordinate $s = -W_{-1}(-t/\alpha)$, which it calls the *upper transition coordinate*; the lower real branch is necessary because $s \rightarrow \infty$, and the branch conventions agree with the DLMF [15]. Its version of Lemma 4.1 (its Lemma 3.1) states $-F'(t) \sim M^2/2$, $B \sim M^3/(\alpha s)$ and $|\kappa_j| = O_{\alpha, j}(M^{j+1}/s)$ for $j \geq 2$, with the following proof.

Second proof of (4.3)–(4.5) (source 48). In the scaled variable $k = Mu$, the log odds equal $\alpha s(1 - u) + \alpha \log u$. For $0 < u < 1$ this tends to $+\infty$, and for $u > 1$ it tends to $-\infty$. Splitting the sum at fixed fractions of M , and using exponential decay above a fixed multiple of M , gives $M^{-2} \sum k p_k \rightarrow \int_0^1 u \, du$.

Near $k = M$, put $v = t(k - M)$. The log odds are

$$-v + \alpha \log \left(1 + \frac{v}{\alpha s} \right) \rightarrow -v.$$

The mesh in v is $t \rightarrow 0$. The derivative estimate used in Lemma 3.6 supplies an exponentially integrable bound on $k \geq M/2$. Thus a Riemann-sum limit gives

$$B \sim \frac{M^2}{t} \int_{-\infty}^{\infty} \frac{e^v}{(1 + e^v)^2} \, dv = \frac{M^2}{t} = \frac{M^3}{\alpha s}.$$

The range below $M/2$ is negligible. To see this for any fixed moment order, split it at $M^{1/4}$ as in Lemma 3.6, now multiplying by k^j . The small range is $O(M^{(j+1)/4})$; the other range is $O_{\alpha}(M^{j+1-c_{\alpha}})$. Above $M/2$, the exponential envelope gives $\sum k^j p_k (1 - p_k) = O_{\alpha, j}(M^{j+1}/s)$. This proves the cumulant bound, and normalization by the variance proves (4.8). $\square$

At the exact saddle, (4.3) gives $M \sim K$ and $s \sim r$. For cumulants, write $b_1(p) = p$ and $b_{j+1}(p) = p(1 - p)b'_j(p)$. Then

$$\kappa_j = \sum_{k \geq 1} k^j b_j(p_k), \quad |b_j(p)| \leq C_j p(1 - p) \quad (j \geq 2). \quad (4.7)$$

Consequently, with $\varepsilon = s/M$,

$$\frac{|\kappa_j|}{B^{j/2}} = O_{\alpha, j}(\varepsilon^{(j-2)/2}), \quad j \geq 3. \quad (4.8)$$

These bounds are deliberately conservative; signed cancellations can make individual corrections smaller. They suffice for all fixed Edgeworth orders.

[Merge note, batch 75.] Sources 75-01 and 75-06 determine those cancellations: $\kappa_j \sim \frac{j!}{2} M^{j+1}/(\alpha s)^{j-1}$, so the normalized cumulants are $O(\varepsilon_{\sharp}^{(j-2)/2})$ with $\varepsilon_{\sharp} = 1/(\alpha M s)$ (Lemma 16.1 and Theorem 16.3).

5 The local limit theorem, including noncentral arcs

5.1 An elementary consecutive-block inequality

Lemma 5.1 (sources 40 and 48). *There is an absolute constant $c_0 > 0$ such that for every integer $h \geq 2$, every integer a , and every $|\theta| \leq \pi$,*

$$\sum_{k=a}^{a+h-1} \sin^2(k\theta/2) \geq c_0 \min(h, h^3\theta^2). \quad (5.1)$$

Proof. The geometric-sum formula gives

$$\sum_{k=a}^{a+h-1} \sin^2(k\theta/2) \geq \frac{1}{2} \left(h - \left| \frac{\sin(h\theta/2)}{\sin(\theta/2)} \right| \right).$$

When $h|\theta| \leq 1$, Taylor bounds for sine show that the difference in parentheses is at least $ch(h^2 - 1)\theta^2$. For $h|\theta| \geq 2\pi$, the bound $|\sin(\theta/2)| \geq |\theta|/\pi$ makes the ratio at most $h/2$. On the intermediate range $1 \leq h|\theta| \leq 2\pi$, its ratio to h is uniformly less than one. To see the uniformity, let $h \rightarrow \infty$ and use uniform convergence to $|\sin(x/2)/(x/2)| < 1$ on $1 \leq x \leq 2\pi$; the finitely many remaining h have the same strict inequality by the strict triangle inequality for a nonconstant geometric progression. Taking the minimum of the resulting positive constants proves the lemma. $\square$

Source 48 states the lemma for every sufficiently large h in the equivalent form $\sum_{k=a}^{a+h-1} \sin^2(k\theta/2) \geq c_0 h \min\{1, h^2\theta^2\}$.

Second proof (source 48). The sum is at least

$$\frac{1}{2} \left(h - \left| \sum_{k=a}^{a+h-1} e^{ik\theta} \right| \right) = \frac{1}{2} \left(h - \frac{|\sin(h\theta/2)|}{|\sin(\theta/2)|} \right).$$

For $h|\theta| \leq 1$, Taylor's formula bounds the deficit below by a constant times $h^3\theta^2$. For $1 \leq h|\theta| \leq C$, divide by h and use the strict inequality $|\sin u| < |u|$ away from zero, uniformly on a compact interval; the denominator converges uniformly to $|h\theta|/2$ in this range. For $h|\theta| \geq C$, use $|\sin(\theta/2)| \geq |\theta|/\pi$, choosing a fixed sufficiently large C . These three ranges prove the result. $\square$

5.2 Fourier decay and the central Gaussian

Let $\phi_t(\theta) = \mathbb{E}_t e^{i\theta(S - \mathbb{E}_t S)}$. The modulus identity for one Bernoulli factor implies

$$|\phi_t(\theta)| \leq \exp \left(-2 \sum_{k \geq 1} p_k(1 - p_k) \sin^2(k\theta/2) \right). \quad (5.2)$$

In a consecutive block of length $h \asymp_\alpha M/s$ centered at M , all $p_k(1 - p_k)$ are bounded below by a positive constant. On $|\theta| \leq c_1/M$, this block gives

$$|\phi_t(\theta)| \leq e^{-c_2 B \theta^2}. \quad (5.3)$$

Indeed $k \asymp M$ on the block, and $\sin(k\theta/2)$ is comparable with $k\theta$ for sufficiently small c_1 . On its complement in $[-\pi, \pi]$, Lemma 5.1 gives

$$|\phi_t(\theta)| \leq \exp(-c_3 M/s^3), \quad c_1/M \leq |\theta| \leq \pi. \quad (5.4)$$

All constants here may depend on α . Notice the power s^3 , rather than s : the random part sizes lie in a relatively narrow consecutive block, so one must control possible near-resonances. An argument checking only a neighborhood of zero would leave a genuine gap. Source 48 proves the same two bounds from the same block (its (5.3) and (5.4) are written with V for B and L for s): the worst case of Lemma 5.1 at $|\theta| = c_1/M$ is a constant times $h^3/M^2 \asymp_\alpha M/s^3$, so in particular no other root-of-unity arc contributes at an algebraic order.

At the saddle, Fourier inversion gives

$$\mathbb{P}_t(S = n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \phi_t(\theta) d\theta.$$

For bounded $x = \theta\sqrt{B}$, Taylor expansion and (4.8) give $\phi_t(x/\sqrt{B}) \rightarrow e^{-x^2/2}$. The central bound dominates this integrand by e^{-cx^2} . The noncentral integral, even after multiplication by $\sqrt{B}$, is $O(\sqrt{B}e^{-c_3M/s^3}) = o(1)$. Thus

$$\mathbb{P}_t(S = n) \sim (2\pi B)^{-1/2}. \quad (5.5)$$

Together with (4.2), this proves the leading part of Theorem 2.2. In particular,

$$\log a_\alpha(n) = F(t_n) + nt_n + O_\alpha(s), \quad (5.6)$$

since $\log B = O(s)$.

5.3 A uniform variant used for conditioning

The same proof gives the following useful extension. Delete either a fixed number of sites $k = O(M)$, or all sites above a cutoff m for which $m/M \rightarrow \gamma > 1$ and the deleted mean and variance are $O(M)$ and $O(M^2)$. The central consecutive block remains available, the retained variance is $B(1 + o(1))$, and the cumulant bounds remain valid. If an integer target differs from the retained mean by $o(\sqrt{B})$, its point probability is

$$(2\pi B)^{-1/2}(1 + o(1)). \quad (5.7)$$

This follows in the central integral by inserting $e^{-ix(m-\mathbb{E}S)/\sqrt{B}} \rightarrow 1$; all the dominating estimates are unchanged. In the fixed-site deletion case, the deleted variance is automatically $O(M^2)$. The assertion is uniform for boundedly many deletions and shifts bounded by a fixed multiple of M . Source 48 records the same extension from its proof of Theorem 2.2, with a factor $e^{-ih\theta}$ in the integral: uniformly for $h = o(\sqrt{B})$,

$$\mathbb{P}_t(S = -F'(t) + h) = \frac{1 + o(1)}{\sqrt{2\pi B}} \quad \text{whenever } -F'(t) + h \text{ is an integer}, \quad (5.8)$$

also after deleting finitely many summands, or after truncating well above the active interval, provided the removed variance is $o(B)$ (source 48).

6 The all-orders coefficient correction

The normalized Fourier integrand has the expansion

$$\phi_t(x/\sqrt{B}) = e^{-x^2/2} \exp\left(\sum_{j \geq 3} \frac{\kappa_j(\mathrm{i}x)^j}{j! B^{j/2}}\right). \quad (6.1)$$

Assign weight $j - 2$ to the j th normalized cumulant. Every monomial of odd weight has odd total power of x and integrates to zero. A monomial of weight 2ℓ has order $O(\varepsilon^\ell)$ by (4.8). Gaussian integration therefore gives

$$\mathcal{E}_\ell = \sum_{\substack{m_3, \dots, m_{2\ell+2} \geq 0 \\ \sum_{j=3}^{2\ell+2} (j-2)m_j = 2\ell}} \frac{(-1)^{T/2} (T-1)!!}{B^{T/2}} \prod_{j=3}^{2\ell+2} \frac{\kappa_j^{m_j}}{(j!)^{m_j} m_j!}, \quad T = \sum_{j=3}^{2\ell+2} j m_j. \quad (6.2)$$

Here T is automatically even. Besides (2.9), the next term is

$$\begin{aligned} \mathcal{E}_2 = & -\frac{\kappa_6}{48B^3} + \frac{7\kappa_3\kappa_5}{48B^4} + \frac{35\kappa_4^2}{384B^4} \\ & - \frac{35\kappa_3^2\kappa_4}{64B^5} + \frac{385\kappa_3^4}{1152B^6}. \end{aligned} \quad (6.3)$$

For completeness, the formal expansion above has the needed integrated remainder. On $|\theta| \leq c/M$, derivatives of order $j \geq 2$ of $\log \mathbb{E}_t e^{i\theta S}$ have absolute value at most $C_{\alpha,j} M^{j+1}/s$. For $k \leq 2M$, take c sufficiently small so that the Bernoulli denominators have modulus comparable with $1 + k^\alpha e^{-tk}$. For $k > 2M$, the odds tend uniformly to zero; the denominators are bounded away from zero and the exponentially decreasing tails give the same bound. These observations justify Taylor's formula with controlled remainders.

Fix J , choose a small $\delta > 0$ depending on J , and first restrict the normalized integral to $|x| \leq \varepsilon^{-\delta}$. Expand through weight $2J - 1$, using Taylor's formula for the exponential. The terms of weight $2J$ and higher have an integrable bound of the form $C_J \varepsilon^J (1 + |x|^{C_J}) e^{-x^2/4}$, after choosing δ small enough. The omitted odd-weight terms integrate to zero on the symmetric interval. Between this interval and $c\sqrt{B}/M$, (5.3) makes the integral smaller than every power of ε . The remaining Fourier range is bounded by (5.4), with the same conclusion. This proves (2.8) and its error for every fixed J .

Remark 6.1 (What an effective finite-size bound would contain). Before absorbing flat terms into the asymptotic O -term, the proof contains an additional relative bound of the shape $C_\alpha \sqrt{B} \exp(-c_\alpha M/s^3)$ for noncentral arcs. This can be much larger than the central Edgeworth corrections at attainable n , especially for larger α . The theorem is not a claim that its first few terms have small relative error uniformly in α or for all moderate n .

6.1 The cumulative tuple form and a second proof (source 48)

Source 48 writes the corrections cumulatively. For a finitely supported tuple $\mathbf{m} = (m_3, m_4, \dots)$ of nonnegative integers put $w(\mathbf{m}) = \sum_{j \geq 3} (j-2)m_j$ and $T(\mathbf{m}) = \sum_{j \geq 3} j m_j$. Then, for every $J \geq 1$,

$$1 + \sum_{\ell=1}^{J-1} \mathcal{E}_\ell = \sum_{\substack{w(\mathbf{m}) < 2J \\ T(\mathbf{m}) \text{ even}}} (-1)^{T(\mathbf{m})/2} (T(\mathbf{m}) - 1)!! \prod_{j \geq 3} \frac{1}{m_j!} \left(\frac{\kappa_j}{j! B^{j/2}} \right)^{m_j}, \quad (6.4)$$

where the empty tuple contributes 1 and $(-1)!! = 1$. This is (6.2) summed over $\ell < J$ (an even T is the same as an even weight), and (2.8) reads $a_\alpha(n) = \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}} ((6.4) + O_{\alpha,J}((s/M)^J))$; the leading formula is a genuine multiplicative equivalent.

Second proof of Theorem 2.2 (source 48). Fourier inversion and (4.2) reduce the assertion to the probability $\mathbb{P}_t(S = n)$ at the exact saddle. Set $u = \theta\sqrt{B}$ and center the characteristic function; its expansion at zero is (6.1), interpreted to any fixed finite Taylor order with a remainder.

The Taylor expansion is uniform for the following reasons. For $|\theta| \leq c_0/M$, derivatives of order $j \geq 2$ of the logarithm of the characteristic function are $O_{\alpha,j}(M^{j+1}/s)$. For k in a fixed multiple of $[1, M]$, either the Bernoulli parameter is away from $1/2$, or the small phase keeps the factor uniformly away from zero. For k above that multiple of M , p_k is small and its factor is again uniformly away from zero. The derivatives in both ranges are bounded by $C_j k^j p_k(1 - p_k)$; summation uses Lemma 4.1. Taylor's integral remainder therefore obeys the corresponding bound from (4.8).

Explicit finite-order bookkeeping for the central integral: fix J , put $D = 2J + 2$, and write

$$Z(u) = \sum_{j=3}^D \frac{\kappa_j(iu)^j}{j! B^{j/2}}.$$

Taylor's remainder in the exponent is bounded by $C_{\alpha,J} \varepsilon^{(D-1)/2} |u|^{D+1}$. Choose $0 < \sigma < 1/6$ sufficiently small that also $\sigma < (D-1)/(2(D+1))$. On $|u| \leq \varepsilon^{-\sigma}$, both $Z(u)$ and this remainder tend uniformly to zero. Consequently replacing the exact exponential by $e^{-u^2/2} e^{Z(u)}$ has integrated error $O_{\alpha,J}(\varepsilon^{J+1/2})$, by the bound $|e^{a+b} - e^a| \leq |b|e^{|a|+|b|}$ and a fixed Gaussian moment.

Next expand e^Z through degree $2J$. Its remainder, integrated against the Gaussian, is again $O_{\alpha,J}(\varepsilon^{J+1/2})$: every coefficient in Z is $O_{\alpha,J}(\varepsilon^{1/2})$, so $|Z|^{2J+1}$ is bounded by that power of ε times a fixed polynomial in $|u|$. In the finite expansion of the retained powers, any monomial of weight at least $2J$ has integrated absolute size $O_{\alpha,J}(\varepsilon^J)$. Every monomial of smaller weight occurs in these powers, because each factor has positive weight and the largest required index is at most $2J + 1$. Odd total degree integrates to zero on the symmetric central interval. The surviving terms are precisely those in (6.4). This argument uses only finite Taylor expansions and their remainders, not uniform convergence of an infinite series.

The remainder of $|\theta| \leq c_0/M$ is smaller than every power of ε by (5.3); the rest of $[-\pi, \pi]$ is likewise negligible by (5.4). Integrating each retained polynomial over the whole real line introduces only another negligible Gaussian tail. The Gaussian moment is zero for odd T and equals $(T-1)!!$ for even T ; the phase i^T supplies the sign. At weight two the only possibilities are $m_4 = 1$ and $m_3 = 2$, giving exactly (2.9). $\square$

7 Continuum smoothing to every logarithmic order

7.1 The inverse at the occupation boundary

Define

$$h_s(u) = s(u-1) - \log u. \quad (7.1)$$

Near $u = 1$ it is strictly increasing. Let $u_s(v)$ be its inverse with $u_s(0) = 1$, and put

$$U_m(s) = u_s^{(m)}(0). \quad (7.2)$$

The derivative operator along the inverse is

$$\frac{d}{dv} = \frac{1}{s - 1/u} \frac{d}{du}. \quad (7.3)$$

Starting with u and applying (7.3) repeatedly, then setting $u = 1$, gives an effective rational-function recursion. Equivalently, Lagrange inversion gives

$$U_m(s) = (m-1)! [z^{m-1}] \left(\frac{z}{sz - \log(1+z)} \right)^m. \quad (7.4)$$

The first required odd derivatives are

$$\begin{aligned} U_1(s) &= \frac{1}{s-1}, \\ U_3(s) &= \frac{2s+1}{(s-1)^5}, \\ U_5(s) &= \frac{24s^3 + 58s^2 + 22s + 1}{(s-1)^9}, \\ U_7(s) &= \frac{720s^5 + 3708s^4 + 4400s^3 + 1452s^2 + 114s + 1}{(s-1)^{13}}. \end{aligned} \quad (7.5)$$

Write $\eta(z) = (1 - 2^{1-z})\zeta(z)$. The elementary even kernel $g(v) = \log(1 + e^{-|v|})$ has moments

$$\int_{\mathbb{R}} v^{2j} g(v) dv = 2(2j)! \eta(2j+2), \quad j \geq 0. \quad (7.6)$$

Indeed expand the logarithm on $(0, \infty)$ and integrate each exponentially decaying term. Odd moments vanish.

Lemma 7.1 (Boundary smoothing expansion). *For every fixed $J \geq 1$,*

$$\int_0^\infty \log(1 + x^\alpha e^{-tx}) dx = M \left(\alpha(s/2 - 1) + \sum_{j=1}^J \frac{2\eta(2j)}{\alpha^{2j-1}} U_{2j-1}(s) + O_{\alpha,J}(s^{-2J-1}) \right), \quad (7.7)$$

where $t = \alpha s e^{-s}$. The expansion may be differentiated with respect to s any fixed number of times, retaining a remainder smaller than any prescribed inverse power of s by taking sufficiently many terms first.

Proof. After $x = Mu$, the integrand is $\log(1 + e^{-\alpha h_s(u)})$. The baseline is $\alpha M \int_0^1 [s(1-u) + \log u] du = \alpha M(s/2 - 1)$. The discrepancy between this signed baseline and the positive-part baseline is confined to the small lower crossing and is exponentially small in s before multiplication by M .

Choose a fixed small $\rho > 0$ and restrict initially to $|u - 1| < \rho$. Replacing the original integrand by its positive-part baseline leaves $g(\alpha h_s(u))$. On a symmetric v -interval $|v| < \gamma s$ inside the image of this neighborhood, changing variables $v = h_s(u)$ gives

$$\int g(\alpha v) u'_s(v) dv.$$

The omitted portions have exponentially small tails. Taylor-expand $u'_s(v)$ through degree $2J - 1$. On the chosen interval, repeated use of (7.3) bounds the derivative of order $2J$ of u'_s by $O_J(s^{-2J-1})$. The Taylor remainder is thus bounded by $C_J s^{-2J-1} |v|^{2J}$, whose product with $g(\alpha v)$ is integrable. After extending the polynomial terms to the whole real line, odd terms vanish, and (7.6) supplies exactly $2\eta(2j)U_{2j-1}(s)/\alpha^{2j-1}$.

Here is a more explicit treatment of the region near zero, which must not be mistaken for part of the main boundary. On $0 < u < e^{-s/2}$, both the original integrand and the absolute baseline have integral at most $C_\alpha s e^{-s/2}$. On $e^{-s/2} \leq u \leq 1 - \rho$, the concavity of $s(1-u) + \log u$ puts it above a positive multiple of s for large s . The smoothing error there is exponentially small. On $u \geq 1 + \rho$ the log odds are negative and decrease at rate comparable with s . All these contributions are $O_{\alpha,J}(s^{-2J-1})$ for every fixed J .

Differentiation with respect to s inserts powers of $1 - u$ and derivatives of the same logistic kernel. On the boundary interval the inverse derivatives still have polynomial bounds in s^{-1} ;

away from it, exponential tail bounds remain after multiplication by any fixed polynomial. Taylor expansion to a higher order therefore gives the differentiated version with any prescribed inverse-power precision. No assumption about differentiating an arbitrary O -term is being used. $\square$

7.2 From the continuum to the lattice

As in Lemma 3.1, the difference between $F(t)$ and its continuum integral is $O_\alpha(s)$. We also need the mean. The function

$$x \mapsto \frac{x^{\alpha+1}e^{-tx}}{1+x^\alpha e^{-tx}} = xp(x)$$

is unimodal and has maximum $O_\alpha(M)$. To verify unimodality, its derivative has the sign of $1+(\alpha-tx)(1-p(x))$. This is positive for $x \leq \alpha/t$. For $x > \alpha/t$, the function $(tx-\alpha)(1-p(x))$ increases strictly from zero to infinity. Thus

$$\sum_{k \geq 1} kp_k = \int_0^\infty xp(x) dx + O_\alpha(M). \quad (7.8)$$

This exact lattice error is negligible at every fixed inverse-logarithmic order relative to the mean M^2 .

Introduce the formal Laurent series at $s = \infty$

$$d_\alpha(s) = \sum_{j \geq 1} \frac{2\eta(2j)}{\alpha^{2j}} U_{2j-1}(s). \quad (7.9)$$

The notation is asymptotic/formal: no convergence of the infinite sum in j is asserted. Each coefficient of s^{-1} uses only finitely many summands, since $U_m(s) = O_m(s^{-m})$. The expansion begins

$$\begin{aligned} d_\alpha(s) = & \frac{c}{s} + \frac{c}{s^2} + \frac{c}{s^3} + \frac{c+7c^2/5}{s^4} + \frac{c+77c^2/10}{s^5} \\ & + \frac{c+49c^2/2+372c^3/35}{s^6} + \dots \end{aligned} \quad (7.10)$$

In the all-orders logarithmic sense,

$$F(t) \sim \alpha M(s/2 - 1 + d_\alpha(s)). \quad (7.11)$$

The lattice error $O(s)$ and the non-boundary continuum pieces are smaller than M/s^R for every fixed R . They are *not* thereby zero and must not be discarded when a multiplicative coefficient equivalent is wanted.

7.3 A second route: an explicit differential operator (source 48)

Source 48 proves the two lattice estimates by the bounded-variation inequality $|\sum_{k \geq 1} f(k) - \int_0^\infty f| \leq \int_0^\infty |f'| = 2 \max f$ for a nonnegative unimodal f with $f(0) = 0$ tending to zero at infinity, applied to $\log(1 + \xi^\alpha e^{-t\xi})$ (maximum $O_\alpha(s)$) and to $\xi p(\xi)$ (maximum $O_\alpha(M)$, unimodal by the argument before (7.8)). It then replaces the inverse u_s and the Lagrange coefficients U_m by one explicit operator. Let

$$f(s) = \frac{s}{s-1}, \quad \mathcal{T} = \frac{s}{s-1} \frac{d}{ds}, \quad (7.12)$$

with η as in Section 7, and define the asymptotic series at $s = \infty$

$$\boxed{\mathcal{C}_\alpha(s) \sim 2 \sum_{j \geq 0} \frac{\eta(2j+2)}{\alpha^{2j+2}} \mathcal{T}^{2j} f(s), \quad \mathcal{B}_\alpha(s) = \mathcal{C}_\alpha(s) + \mathcal{T} \mathcal{C}_\alpha(s).} \quad (7.13)$$

The formula is locally finite after expansion in s^{-1} : for $j \geq 1$, $\mathcal{T}^{2j} f(s) = O(s^{-2j-1})$, so only finitely many summands affect any prescribed coefficient.

Lemma 7.2 (upper-transition expansion; source 48). *With $t = \alpha s e^{-s}$, in the sense of asymptotic series to every fixed finite order,*

$$F(t) \sim \frac{\alpha^2}{t} \left(\frac{s^2}{2} - s + \mathcal{C}_\alpha(s) \right), \quad (7.14)$$

$$-F'(t) \sim \frac{\alpha^2}{t^2} \left(\frac{s^2}{2} + \mathcal{B}_\alpha(s) \right). \quad (7.15)$$

More precisely, replacing $\mathcal{C}_\alpha$ or $\mathcal{B}_\alpha$ by its polynomial through s^{-R} leaves $O_{\alpha,R}(s^{-R-1})$ inside the corresponding parentheses, for every fixed $R \geq 0$.

Since $\alpha^2/t = \alpha M/s$, (7.14) is (7.11) with $d_\alpha(s) = \mathcal{C}_\alpha(s)/s$, and (7.15) is the mean relation (8.1) with $Q_\alpha(s) = 1 + 2\mathcal{B}_\alpha(s)/s^2$.

Proof. In the continuum integral of F , with integration variable ξ , put $y = t\xi/\alpha$. Since $\log(\alpha/t) = s - \log s$, the exponent becomes

$$\alpha \log \xi - t\xi = \alpha \tilde{h}_s(y), \quad \tilde{h}_s(y) = s - y + \log(y/s).$$

The zero-temperature contribution is

$$\int_0^s \alpha \tilde{h}_s(y) dy = \alpha(s^2/2 - s). \quad (7.16)$$

Subtracting this reference integral is legitimate even though $\tilde{h}_s$ becomes negative very near zero; that region is included in the error estimate.

On $y \geq s/2$, use the increasing coordinate $v = y - s - \log(y/s) = -\tilde{h}_s(y)$. If $y = y(v)$ is its inverse, then $y(0) = s$ and $y'(v) = y(v)/(y(v) - 1) = f(y(v))$. The difference from the reference integrand is $g(\alpha v) = \log(1 + e^{-\alpha|v|})$, an even exponentially decreasing function, with moments

$$\int_{-\infty}^{\infty} v^{2j} g(\alpha v) dv = \frac{2(2j)!}{\alpha^{2j+1}} \eta(2j+2), \quad (7.17)$$

by (7.6); these are the elementary Fermi–Dirac moments underlying Sommerfeld expansions [16]. Since $d/dv = f(y) d/dy$, Taylor expansion of $y'(v)$ at zero gives

$$\left. \frac{d^{2j}}{dv^{2j}} y'(v) \right|_{v=0} = \mathcal{T}^{2j} f(s).$$

On $|v| \leq s/4$, $y(v) \asymp s$, and for each fixed $a \geq 1$, $\mathcal{T}^a f(y) = O_a(s^{-a-1})$. Expanding through the odd degree $2J+1$, integrating against the even kernel, and bounding the next derivative gives an error $O_{\alpha,J}(s^{-2J-3})$. The portions with $|v| > s/4$ are exponentially small, since $y'(v)$ stays bounded on the domain $y \geq s/2$.

The range $0 < y < s/2$ also has exponentially small error *inside the scaled integral*. Split it at $e^{-s/2}$. On the small interval, integrate the bound $C_\alpha(1 + s + |\log y|)$, obtaining $O_\alpha(s e^{-s/2})$. On

$[e^{-s/2}, s/2]$, concavity gives $\tilde{h}_s(y) \geq s/2 - \log s - O(1)$, so the discrepancy is $O_\alpha(s^{\alpha+1}e^{-\alpha s/2})$. These errors need not remain exponentially small in the original, unscaled variable, but they are smaller than every inverse power of s inside the present parentheses. Multiplying the integral by α/t , the moments (7.17) give (7.14); the sum–integral error is absorbed at every fixed order.

The mean formula is proved directly, rather than by differentiating an unspecified big- O term. The scaled mean integral is $\frac{\alpha^2}{t^2} \int_0^\infty y (1 + e^{\alpha y})^{-1} dy$, with reference integral $\int_0^s y dy = s^2/2$. Under the same coordinate change the difference kernel is the odd function

$$\mathcal{K}_\alpha(v) = \frac{1}{1 + e^{\alpha v}} - \mathbf{1}_{v < 0},$$

integrated against $y(v)y'(v)$. Its odd moments are

$$\int_{-\infty}^\infty v^{2j+1} \mathcal{K}_\alpha(v) dv = \frac{2(2j+1)!}{\alpha^{2j+2}} \eta(2j+2).$$

The elementary operator identity $\mathcal{T}(sf(s)) = f(s) + \mathcal{T}f(s)$ shows that the resulting correction is precisely $\mathcal{C}_\alpha + \mathcal{T}\mathcal{C}_\alpha$. The same cutoff and Taylor estimates apply, and (7.8) absorbs the discretization error. This proves (7.15) with controlled remainders, independently of any differentiation rule for errors. $\square$

In the parameter c , the beginning of (7.13) is

$$\mathcal{C}_\alpha(s) = c + \frac{c}{s} + \frac{c}{s^2} + \frac{c + \frac{7}{5}c^2}{s^3} + \frac{c + \frac{77}{10}c^2}{s^4} + \frac{c + \frac{49}{2}c^2 + \frac{372}{35}c^3}{s^5} + O_\alpha(s^{-6}), \quad (7.18)$$

$$\mathcal{B}_\alpha(s) = c + \frac{c}{s} + \frac{-2c + \frac{7}{5}c^2}{s^3} + \frac{-5c + \frac{7}{2}c^2}{s^4} + O_\alpha(s^{-5}). \quad (7.19)$$

In particular, the coefficient of s^{-2} in $\mathcal{B}_\alpha$ is zero, and $\mathcal{C}_\alpha(s)/s$ reproduces (7.10) term by term.

[Merge note, batch 7301.] The intake recomputed (7.18), (7.19), (8.8) and $P_1, \dots, P_6$ from (7.13) and (8.10) in exact rational arithmetic, for three values of c ; all agree.

8 Eliminating the saddle and generating all coefficients

Since $dt/ds = -\alpha(s-1)/M$, differentiating the continuum expansion and using (7.8) gives the formal mean relation

$$n \sim \frac{M^2}{2} Q_\alpha(s), \quad Q_\alpha(s) = 1 + \frac{2(d_\alpha(s) + d'_\alpha(s))}{s-1}. \quad (8.1)$$

Here d'_α denotes formal differentiation. Thus

$$r = s + \frac{1}{2} \log Q_\alpha(s). \quad (8.2)$$

Because $Q_\alpha(s) = 1 + O(s^{-2})$, this has a unique formal inverse $s = r + O(r^{-2})$. It is computed recursively, for example by iterating $s \leftarrow r - \frac{1}{2} \log Q_\alpha(s)$ in the ring of formal series in r^{-1} . Every iteration determines additional coefficients.

Substitution in $F(t) + nt$ gives

$$\mathcal{H}_\alpha(r) = \frac{s(Q_\alpha(s) + 1)/2 - 1 + d_\alpha(s)}{\sqrt{Q_\alpha(s)}}, \quad r = s + \frac{1}{2} \log Q_\alpha(s). \quad (8.3)$$

By construction,

$$\mathcal{H}_\alpha(r) = r - 1 + \sum_{j \geq 1} P_j(c) r^{-j}. \quad (8.4)$$

Equations (7.3), (7.9), (8.1), and (8.3) are an all-orders algorithm involving only rational operations and expansions of logarithms and square roots. Since even zeta values are rational multiples of powers of π^2 , the coefficients are polynomials over $\mathbb{Q}$ in c .

Proof of Theorem 2.1. Fix the desired order R . Apply Lemma 7.1 and its mean version with more than enough terms that both remainders, after all operations, are $O_{\alpha,R}(r^{-R-1})$ in the normalized entropy. The exact mean relation has an additional error $O(M)$, negligible compared with $M^2 r^{-A}$ for every fixed A . Inverting (8.2) is stable because its derivative is $1 + O(s^{-3})$. Finite Taylor substitution is consequently legitimate with remainders, not only as a formal manipulation.

We obtain

$$F(t_n) + nt_n = \alpha K \left(r - 1 + \sum_{j=1}^R P_j(c) r^{-j} + O_{\alpha,R}(r^{-R-1}) \right).$$

Equation (5.6) contributes only $O(s)$, which is $o(Kr^{-R-1})$. This proves (2.2). Direct expansion gives (2.3); the supplied symbolic script reproduces those coefficients. $\square$

One further coefficient for an independent check. The same algorithm gives

$$P_8(c) = c + \frac{259}{2}c^2 + \frac{7271}{10}c^3 + \frac{157813}{840}c^4. \quad (8.5)$$

The continuum quadrature script checks the approach of the remainder after P_7 , multiplied by r^8 , to this polynomial. It does not replace the preceding proof or evaluate integer coefficients at those enormous sizes.

[Merge note, batch 75.] Source 75-06 prints $b_8(u) = u(157813u^3 + 3664584u^2 + 3916080u + 181440)/1088640$, which equals (8.5) at $u = 6c$, and its `code/75-06-powers-formal.py` derives it by exact arithmetic over $\mathbb{Q}[u]$. So P_8 is now also confirmed exactly, independently of source 40's algorithm, not only by the continuum diagnostic.

8.1 The same elimination in operator form (source 48)

At the exact saddle, Lemma 7.2 gives, to every finite inverse-logarithmic order,

$$K \sim M \left(1 + \frac{2\mathcal{B}_\alpha(s)}{s^2} \right)^{1/2}, \quad (8.6)$$

meaning the full finite-order expansion, not merely a leading ratio. Equivalently, its formal inverse is

$$r = s + \frac{1}{2} \log \left(1 + \frac{2\mathcal{B}_\alpha(s)}{s^2} \right), \quad (8.7)$$

which is (8.2). There is a unique solution with $s = r + O(r^{-2})$ as a formal series in r^{-1} . Indeed, put $z = r^{-1}$ and $\nu = s - r$. Then $s^{-1} = z/(1 + \nu z)$. The right-hand side of the fixed-point equation

$$\nu = -\frac{1}{2} \log \left(1 + \frac{2\mathcal{B}_\alpha(r + \nu)}{(r + \nu)^2} \right)$$

has order z^2 , and changing ν by $O(z^d)$ changes the right-hand side by $O(z^{d+3})$. Iteration starting with $\nu = 0$ therefore determines every coefficient after finitely many steps. The first terms are

$$s = r - \frac{c}{r^2} - \frac{c}{r^3} + \frac{c^2}{r^4} + \frac{2c - \frac{7}{5}c^2}{r^5} + O_\alpha(r^{-6}). \quad (8.8)$$

At the saddle, addition of (7.14) and nt gives

$$F(t) + nt \sim \alpha M \left(s - 1 + \frac{2\mathcal{C}_\alpha(s) + \mathcal{T}\mathcal{C}_\alpha(s)}{s} \right). \quad (8.9)$$

The logarithm of the Gaussian prefactor in Theorem 2.2 is $O_\alpha(s)$, and the logarithm of its relative correction is $o(1)$; both are smaller than every scale K/r^R . Thus the formal series determining Theorem 2.1 is

$$\boxed{\mathcal{H}_\alpha(r) = \frac{s - 1 + (2\mathcal{C}_\alpha(s) + \mathcal{T}\mathcal{C}_\alpha(s))/s}{\sqrt{1 + 2\mathcal{B}_\alpha(s)/s^2}}, \quad \mathcal{H}_\alpha(r) = r - 1 + \sum_{j \geq 1} P_j(c)r^{-j}}, \quad (8.10)$$

with s substituted from (8.7). This is (8.3): with $d_\alpha = \mathcal{C}_\alpha/s$ and $Q_\alpha = 1 + 2\mathcal{B}_\alpha/s^2$ the numerators agree, since $s(Q_\alpha + 1)/2 - 1 + d_\alpha = s - 1 + (\mathcal{B}_\alpha + \mathcal{C}_\alpha)/s$.

Why the formal rule proves actual asymptotics. Fix a desired final order R . Take sufficiently many terms, for example through order s^{-R-4} , in the two estimates of Lemma 7.2. Their remainders are genuine uniform estimates on large real s . The derivative of the finite approximation in (8.7) is $1 + O(s^{-3})$, bounded away from zero. The mean value theorem transports its error to an error of the same inverse-log order. Substitution into (8.9), whose normalized derivative is bounded at the relevant order, then gives the remainder in (2.2). This is a finite calculation with controlled remainders, not an assumption that the infinite formal series converges. Together with Theorem 2.2, this is source 48's proof of Theorem 2.1; substitution of (7.18)–(8.8) into (8.10) gives $P_1, \dots, P_6$.

No special symbolic integration routine is needed. In the variable $z = s^{-1}$, the operator is

$$\mathcal{T} = -\frac{z^2}{1-z} \frac{d}{dz}, \quad f = \frac{1}{1-z}. \quad (8.11)$$

The factors $2\eta(2j+2)/\alpha^{2j+2}$ are rational multiples of c^{j+1} , since even zeta values are rational multiples of even powers of π . Truncated polynomial arithmetic with (8.11), followed by the fixed-point substitution, generates the coefficients at any prescribed finite order. The program `code/48-a022629-coefficients.py` implements this construction and verifies the six displayed polynomials and six inverse coefficients exactly.

8.2 An entropy factor beyond the largest summand (source 48)

Combining Theorem 2.1 with Proposition 3.5 gives an additional conclusion.

Corollary 8.1 (source 48).

$$\log \frac{a_\alpha(n)}{\Pi_\alpha(n)} \sim \frac{\pi^2}{6\alpha} \frac{K}{r}. \quad (8.12)$$

Thus the total weighted enumeration exceeds its largest individual weight by a factor that tends to infinity, and whose logarithm has a precise leading constant.

For $\alpha = 1$, this explains why using only the most heavily weighted partition gets the first two large terms right but misses a multiplicatively enormous factor. The transition contains approximately M/s available sizes, and its accumulated smoothing contribution produces this factor.

9 Lambert- W inversion with an error theorem

The coefficients are strictly increasing for $n \geq 1$. Indeed, increasing the largest part of a distinct partition by one is an injection into distinct partitions of $n + 1$, and strictly increases each positive norm weight. Thus the threshold index $N_\alpha(Y)$ in Theorem 2.3 is well-defined for large Y .

The dominant inversion problem is

$$Y = \alpha K(\log K - 1).$$

Writing $w = \log K - 1$ gives $we^w = Y/(\alpha e)$, proving (2.10). In particular $\log K_0 = R_0$. Set $K = K_0 E$. The all-orders formal inversion is the single equation

$$E \mathcal{H}_\alpha(R_0 + \log E) = R_0 - 1, \quad E = 1 + O(R_0^{-2}). \quad (9.1)$$

The derivative with respect to E is $R_0 + O(1)$ at $E = 1$, so successive coefficients are uniquely determined. In terms of c , they are

$$\begin{aligned} E = 1 - \frac{c}{R_0^2} - \frac{c}{R_0^3} + \frac{3c^2/2 - c}{R_0^4} - \frac{c + 2c^2/5}{R_0^5} \\ - \frac{c + 46c^2/5 + 5c^3/2}{R_0^6} + \cdots, \end{aligned} \quad (9.2)$$

and hence

$$\begin{aligned} E^2 = 1 - \frac{2c}{R_0^2} - \frac{2c}{R_0^3} + \frac{4c^2 - 2c}{R_0^4} + \frac{6c^2/5 - 2c}{R_0^5} \\ - \frac{2c + 77c^2/5 + 8c^3}{R_0^6} + \cdots. \end{aligned} \quad (9.3)$$

The asymptotic index is $K_0^2 E^2/2$.

To justify inversion of the sequence rather than just of a formal expression, fix an order, use Theorem 2.1 two orders farther than needed, and bracket the target by the corresponding smooth functions with the two signs of its error bound. Their derivatives with respect to K are $\alpha r(1 + o(1)) > 0$. An entropy error $O(Kr^{-R-1})$ therefore changes K by $O(Kr^{-R-2})$, and changes $n = K^2/2$ by $O(K^2 r^{-R-2})$. These bounds transfer the formal inversion to the actual threshold by monotonicity. Rounding the brackets to integers costs $O(1)$, smaller than all of these fixed-order error scales. This proves Theorem 2.3 and its all-orders extension.

Remark 9.1 (Two different Lambert branches). The saddle phase uses W_{-1} because $s \rightarrow \infty$ solves $se^{-s} = t/\alpha \rightarrow 0^+$. The threshold inverse uses W_0 at a positive argument. Interchanging the branches would destroy the relevant scale. If inversion in elementary logarithms is desired, the usual formal solution of $w + \log w = \log(Y/(\alpha e))$ can be substituted into (9.3); retaining W_0 is shorter and preserves more precision.

9.1 The same inversion in source 48's form (source 48)

Source 48 proves Theorem 2.3 independently (its Theorem 6.1), with the threshold written in the logarithm, $N_\alpha(Y) = \min\{n \geq 1 : \log a_\alpha(n) \geq Y\}$; for a threshold on the sequence itself one uses $N_\alpha(\log \cdot)$. The minimum is well defined because the coefficients diverge (Proposition 3.2) and increase strictly for $n \geq 1$ by the injection above; the initial equality $a_\alpha(0) = a_\alpha(1) = 1$ is

immaterial. Its statement displays two more orders than (2.11):

$$N_\alpha(Y) = \frac{K_0^2}{2} \left(1 - \frac{2c}{R_0^2} - \frac{2c}{R_0^3} + \frac{4c^2 - 2c}{R_0^4} + \frac{\frac{6}{5}c^2 - 2c}{R_0^5} - \frac{8c^3 + \frac{77}{5}c^2 + 2c}{R_0^6} + O_\alpha(R_0^{-7}) \right), \quad (9.4)$$

the coefficients of (9.3); integer rounding changes the result by at most an additional $O(1)$, absorbed in every fixed inverse-logarithmic remainder. With $E = 1 + \delta$ and $C(r) = \sum_{j \geq 1} P_j(c)r^{-j}$, its normalized equation for the corrected inverse,

$$\delta = - \frac{(1 + \delta) \log(1 + \delta) - \delta + (1 + \delta)C(R_0 + \log(1 + \delta))}{R_0}, \quad (9.5)$$

is (9.1) solved for δ . Starting from $\delta = 0$, finite formal iteration gives the unique series with $\delta = O(R_0^{-2})$, whose coefficients are those of $E - 1$ in (9.2).

Second proof of the threshold transfer (source 48). The core equation $K(\log K - 1) = Y/\alpha$ is solved exactly by $K = K_0$, using $we^w = Y/(\alpha e)$. This is the principal, positive Lambert branch, distinct from the lower branch used for the saddle. Choose any finite truncation of Theorem 2.1 and define its continuous approximation on large real K . Its derivative is $\alpha(\log K + O_\alpha(1/\log K))$, positive and comparable to $\alpha \log K$. An error $O_{\alpha,R}(Kr^{-R-1})$ in the logarithm therefore moves its inverse by $O_{\alpha,R}(Kr^{-R-2})$. This follows directly from the mean value theorem on a fixed relative neighborhood of the core root; the preliminary leading asymptotic puts the true root in that neighborhood. For the discrete threshold, apply these upper and lower approximations at $N_\alpha(Y)$ and $N_\alpha(Y) - 1$. Adjacent values of $K = \sqrt{2n}$ differ by $O(1/K)$, so the final bracketing introduces only an $O(1)$ uncertainty in n . Taking the forward expansion to sufficiently high fixed order proves any desired inverse order. $\square$

The first inverse correction is negative: the actual enumeration is larger than its core approximation, so it reaches a specified threshold sooner. For A022629 ($\alpha = 1$, $2c = \pi^2/3$) this begins

$$N_1(Y) = \frac{1}{2} \left(\frac{Y}{W_0(Y/e)} \right)^2 \left(1 - \frac{\pi^2}{3(1 + W_0(Y/e))^2} + O((1 + W_0(Y/e))^{-3}) \right). \quad (9.6)$$

The distinction between inversion in $Y = \log a(n)$ and inversion in $a(n)$ is essential; replacing Y by the original threshold would give a wholly wrong scale (source 48).

[Merge note, batch 75.] Source 75-01 proves Theorem 2.3 through R_0^{-5} and source 75-06 through R_0^{-8} , for μ layers; the coefficients of R_0^{-7} and R_0^{-8} in (9.3) are Δ_7 and Δ_8 of Table 5. The core equation is an instance of the dominant-block theorem of the transseries-and-inversion volume [25], and Theorem 16.4 replaces the $O(1)$ rounding bracket by a $1 + o(1)$ localization from the exact saddle.

10 Conditioned geometry: a logistic boundary and a Gumbel maximum

Conditioning the Bernoulli law (4.1) on $S = n$ gives exactly

$$\mathbb{P}_n(\lambda) = \frac{N(\lambda)^\alpha}{a_\alpha(n)}. \quad (10.1)$$

No approximation is involved in this identification.

10.1 The boundary profile

Fix distinct integer sites $k_{1,n}, \dots, k_{d,n}$ with $(k_{j,n} - M)/H \rightarrow x_j \in \mathbb{R}$. Their unconditioned probabilities tend to $p(x_j) = (1 + e^{x_j})^{-1}$ by Lemma 4.1. For any prescribed pattern $\epsilon_j \in \{0, 1\}$, independence gives

$$\begin{aligned} & \mathbb{P}_n(X_{k_{j,n}} = \epsilon_j \text{ for all } j) \\ &= \prod_{j=1}^d p_{k_{j,n}}^{\epsilon_j} (1 - p_{k_{j,n}})^{1-\epsilon_j} \frac{\mathbb{P}_t\{S^{\text{rem}} = n - \sum_j k_{j,n} \epsilon_j\}}{\mathbb{P}_t(S = n)}. \end{aligned} \quad (10.2)$$

The shifted target differs from the remaining mean by $O(M) = o(\sqrt{B})$. The uniform local limit theorem in Section 5.3 makes the ratio of point probabilities tend to one. This proves both the logistic limit and finite-site asymptotic independence. Similarly, a site at γM has limiting occupation 1 for $0 < \gamma < 1$, and 0 for $\gamma > 1$.

10.2 The correct centering of the maximum

The large solution of (2.12) exists uniquely for sufficiently small t . With $z = tb$, the equation is $z^\alpha e^{-z} = t^{\alpha+1}$, so

$$b(t) = -\frac{\alpha}{t} W_{-1} \left(-\frac{t^{1+1/\alpha}}{\alpha} \right). \quad (10.3)$$

Solving its logarithm, $z - \alpha \log z = -(\alpha + 1) \log t$, with $t = \alpha s e^{-s}$ yields

$$\frac{b}{M} = 1 + \frac{1}{\alpha} + \frac{-\log s + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha}{\alpha s} + O_\alpha \left(\frac{\log s}{s^2} \right). \quad (10.4)$$

Keeping only $b \sim (1 + 1/\alpha)M$ is not adequate for the Gumbel centering: the omitted shift has size $M \log s/s$, larger than $1/t$. Keeping all the displayed terms in (10.4) does give an $o(1/t)$ centering error. The exact definition (2.12) or (10.3) avoids this truncation bookkeeping altogether.

Proof of the maximum assertion in Theorem 2.4. Fix $x \in \mathbb{R}$ and put $m = \lfloor b + x/t \rfloor$. Then $tm \rightarrow \infty$ and $m/M \rightarrow 1 + 1/\alpha > 1$. In the tail $k > m$, all odds tend uniformly to zero. A sum–integral comparison followed by integration by parts gives

$$\sum_{k>m} p_k \sim \sum_{k>m} k^\alpha e^{-tk} \sim \frac{m^\alpha e^{-tm}}{t} \longrightarrow e^{-x}. \quad (10.5)$$

For the second equivalence, substitute $v = tx$ in the integral and use $\int_z^\infty v^\alpha e^{-v} dv = z^\alpha e^{-z}(1 + O_\alpha(z^{-1}))$. The endpoint discretization has relative size $O(t)$. The last limit follows from $b^\alpha e^{-tb}/t = 1$ and $m/b \rightarrow 1$. Also, $\sup_{k>m} p_k = O(t)$, so $\sum_{k>m} p_k^2 = o(1)$. Therefore the unconditioned probability of no part above m is

$$\prod_{k>m} (1 - p_k) \longrightarrow e^{-e^{-x}}. \quad (10.6)$$

It remains to justify conditioning, rather than silently replacing one ensemble by another. The removed mean and variance satisfy

$$\sum_{k>m} k p_k = O(M), \quad \sum_{k>m} k^2 p_k (1 - p_k) = O(M^2).$$

These bounds follow from the same tail-integral calculation with one or two extra powers of k . The remaining sum $S_{\leq m}$ thus has mean $n - O(M)$ and variance $B(1 + o(1))$. Its central block around M is intact. The uniform local limit estimate gives

$$\frac{\mathbb{P}_t(S_{\leq m} = n)}{\mathbb{P}_t(S = n)} \longrightarrow 1.$$

Independence of the retained and removed sites now gives

$$\mathbb{P}_n(L_n \leq m) = \prod_{k > m} (1 - p_k) \frac{\mathbb{P}_t(S_{\leq m} = n)}{\mathbb{P}_t(S = n)} \longrightarrow e^{-e^{-x}}.$$

This proves the Gumbel limit. Its tightness, (10.4), $M \sim K$, and $(tM)^{-1} = 1/(\alpha s) \rightarrow 0$ imply $L_n/K \rightarrow 1 + 1/\alpha$ in probability. $\square$

The two locations are genuinely different. The occupation boundary lies near $M \sim \sqrt{2n}$, while the maximum lies near $(1 + 1/\alpha)M$. For $\alpha = 1$, a typical largest part is asymptotic to $2\sqrt{2n}$, not to $\sqrt{2n}$. Individual occupancies far above the boundary are small, but there are enough possible sites to move the extreme by a macroscopic multiple of M . This is why a boundary profile alone does not determine the largest part.

10.3 Explicit centring in $\log \sqrt{2n}$ (source 48)

Source 48 proves the Gumbel law independently, with the centring written in $K = \sqrt{2n}$ and $r = \log K$ instead of the exact root $b(t_n)$. For positive integer α the measure $\mathbb{P}_n$ is the size distribution obtained by choosing uniformly among the coloured objects; for every $t > 0$ it is exactly the conditional distribution of $\{k : X_k = 1\}$ given $S = n$ (source 48).

Corollary 10.1 (Gumbel law with explicit centring; sources 48 and 40). *Define*

$$\beta_\alpha(r) = (\alpha + 1)r - \log r + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha. \quad (10.7)$$

For every fixed real x ,

$$\mathbb{P}_n \left(\frac{\alpha r}{K} L_n - \beta_\alpha(r) \leq x \right) \longrightarrow \exp(-e^{-x}). \quad (10.8)$$

For A022629 ($\alpha = 1$), an equivalent form is

$$\frac{L_n - \left(2K - \frac{K}{r} \log(r/2) \right)}{K/r} \implies G, \quad \mathbb{P}(G \leq x) = e^{-e^{-x}}. \quad (10.9)$$

Given Theorem 2.4, the corollary amounts to $t_n b(t_n) - \beta_\alpha(r) \rightarrow 0$: by (10.4), $t_n b = \alpha s b/M$ agrees with $(\alpha + 1)s - \log s + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha$ up to $O_\alpha(\log s/s)$, and $s = r + O(r^{-2})$, $t_n = (\alpha r/K)(1 + O(r^{-2}))$ by (8.6) and (8.8). (The intake compared the two centrings numerically for $\alpha = 1, 2, 5$; the difference decays like $\log s/s$.) Source 48 proves it directly.

Proof (source 48). Work first with the unconditioned Bernoulli variables at the exact saddle, with s, M, t as in (2.7). Set

$$d = (\alpha + 1)s - \log s + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha, \quad m = \left\lfloor \frac{d + x}{t} \right\rfloor.$$

Then $tm \rightarrow \infty$ and $m/M \rightarrow (\alpha + 1)/\alpha > 1$. Summation versus integration for a decreasing tail, followed by one integration by parts (or its standard real-power analogue), gives

$$\sum_{k>m} k^\alpha e^{-tk} \sim t^{-\alpha-1} (tm)^\alpha e^{-tm}. \quad (10.10)$$

For real $\alpha > 0$, the relative error from the integral asymptotic is $O_\alpha((tm)^{-1})$; the relative discretization error is $O_\alpha(t)$. Now $\log(1/t) = s - \log s - \log \alpha$ and $\log(d+x) = \log s + \log(\alpha + 1) + o(1)$. Hence

$$(\alpha + 1) \log(1/t) + \alpha \log(tm) - tm = -x + o(1),$$

and the tail in (10.10) tends to e^{-x} . The maximum individual odds on this tail are $O_\alpha(t)$ and tend to zero, so replacing odds by p_k changes the sum by $o(1)$, and $\sum_{k>m} p_k^2 = o(1)$. Independence gives

$$\mathbb{P}_t(X_k = 0 \text{ for every } k > m) = \prod_{k>m} (1 - p_k) \longrightarrow e^{-e^{-x}}.$$

It remains to justify conditioning on total size, which cannot be omitted. Let $T = \sum_{k>m} kX_k$ and $S^{\leq m} = S - T$. The same tail calculation with extra factors of k gives $\mathbb{E}_t T = O_\alpha(M)$ and $\text{Var}_t(T) = O_\alpha(M^2)$. By Lemma 4.1, $\mathbb{E}_t T/\sqrt{B} \rightarrow 0$ and $\text{Var}_t(T)/B \rightarrow 0$. The entire active interval near M lies below m for large n . Thus the shifted local limit estimate (5.8) applies to the truncated sum, and $\mathbb{P}_t(S^{\leq m} = n)/\mathbb{P}_t(S = n) \rightarrow 1$. Independence of the tail and the truncated sum now gives

$$\mathbb{P}_n(L_n \leq m) = \mathbb{P}_t(T = 0) \frac{\mathbb{P}_t(S^{\leq m} = n)}{\mathbb{P}_t(S = n)} \longrightarrow e^{-e^{-x}}.$$

This proves the result in the exact saddle coordinates. Finally, (8.6) and (8.8) imply $M/K = 1 + O_\alpha(r^{-2})$, $s = r + O_\alpha(r^{-2})$, and $t = (\alpha r/K)(1 + O_\alpha(r^{-2}))$. The change in the normalized extreme-value coordinate is $o(1)$, since L_n/M is tight by the limit just proved. Replacing s, M, t by $r, K, \alpha r/K$ therefore proves (10.8); tightness of the Gumbel coordinate also gives $L_n/K \rightarrow 1 + 1/\alpha$. For $\alpha = 1$, $\beta_1(r) = 2r - \log(r/2)$, which is (10.9). $\square$

The upper transition is near $M \sim K$, but the largest part is near $(1 + 1/\alpha)K$. There is no contradiction: a long tail of individually rare occupancies produces an extreme well beyond the transition (source 48).

Source 48 also proves the logistic part of Theorem 2.4 (its Proposition 7.2): for fixed distinct reals $u_1, \dots, u_d$ and $k_j = \lfloor M + u_j/t \rfloor$ at the exact saddle, the inclusion indicators converge jointly to independent Bernoulli variables with probabilities $(1 + e^{u_j})^{-1}$. Since $tH = s/(s-1) \rightarrow 1$, its coordinate $u = t(k - M)$ and source 40's $x = (k - M)/H$ have the same limit, and its proof is the argument around (10.2): the log odds at k_j are $-u_j + o(1)$, and fixing finitely many indicators moves the remaining target by $O_\alpha(M)$ and removes variance $O_\alpha(M^2)$, both negligible.

[Merge note, batch 75.] Source 75-06 proves both parts of Theorem 2.4 independently, for any number μ of layers (Theorem 18.4); at $\mu = 1$ its centring equation is (2.12).

11 A proved exponential scale from Fourier resonances

The bound (5.4) suggests an exponentially small scale not visible in either the logarithmic block or the central Edgeworth series. We can identify its constant at each fixed integer resonance.

Theorem 11.1 (Fixed-resonance modulus). *For every fixed nonzero integer ℓ , with ϕ_t as in Section 5,*

$$-\log \left| \phi_t \left(\frac{2\pi\ell}{M} \right) \right| \sim \frac{2\pi^4\ell^2}{3} \frac{H^3}{M^2} \sim \frac{2\pi^4\ell^2}{3\alpha^3} \frac{M}{s^3}. \quad (11.1)$$

Proof. The exact modulus formula is

$$\log |\phi_t(\theta)| = \frac{1}{2} \sum_{k \geq 1} \log \left(1 - 4p_k(1 - p_k) \sin^2(k\theta/2) \right).$$

At $\theta = 2\pi\ell/M$, write $k = M + xH$. For bounded x ,

$$\sin^2(\pi\ell k/M) = \pi^2\ell^2 x^2 H^2/M^2 + O_\ell(x^4 H^4/M^4).$$

The logarithm is therefore $-4p_k(1 - p_k)$ times this sine square to leading order. Lemma 4.1, applied with moments 2 and 4, gives

$$-\log |\phi_t(2\pi\ell/M)| \sim \frac{2\pi^2\ell^2 H^3}{M^2} \int_{\mathbb{R}} x^2 \frac{e^x}{(1 + e^x)^2} dx.$$

One may first restrict to $|x| \leq s^\delta$ with small fixed $\delta > 0$; Taylor remainders are then uniformly controlled. Outside this interval, the exponential boundary tails and the lower-endpoint bounds in Lemma 4.1 are negligible at the displayed scale. The quadratic logarithm error is $O(H^5/M^4) = o(H^3/M^2)$. Finally,

$$\int_{\mathbb{R}} x^2 \frac{e^x}{(1 + e^x)^2} dx = 4 \sum_{j \geq 1} \frac{(-1)^{j-1}}{j^2} = \frac{\pi^2}{3},$$

by expansion on the positive half-line. This proves the theorem. $\square$

For fixed α , the scale $\exp[-C_\alpha M/s^3]$ is smaller than every power of s/M , because M/s^3 dominates $\log M$. However, its constant deteriorates as α^{-3} . This makes the fixed- α asymptotic theorem compatible with large visible resonances at moderate n .

The theorem concerns the *modulus at a specified point*. It does not yet determine the shifted secondary saddles, their phases, or the contribution of their full neighborhoods to the coefficient integral. Calling it a complete multi-saddle transseries would be incorrect. It does identify a rigorous exponential scale and a concrete next target for exponential improvement.

12 Reproducible computation and finite-size diagnostics

12.1 Exact calculations

The supplied `code/40-norm-moments-verify.py` computes the five integer sequences through $n = 5000$ by descending polynomial multiplication. It compares their initial sixteen terms against the consulted OEIS entries and independently checks $n \leq 150$ by a logarithmic-derivative recurrence. Explicitly, define

$$C_\alpha(m) = \sum_{d|m} (-1)^{m/d+1} d^{1+\alpha m/d}. \quad (12.1)$$

Then $a_\alpha(0) = 1$ and

$$na_\alpha(n) = \sum_{m=1}^n C_\alpha(m) a_\alpha(n-m). \quad (12.2)$$

This follows by expanding $qA'_\alpha(q)/A_\alpha(q)$ formally. The two algorithms have different arithmetic paths, so their agreement is a useful implementation check. It is not a proof of the asymptotic theorems.

All five exact prefix checks, all recurrence checks through 150, and strict monotonicity checks through 5000 passed. The complete computed lists are included as plain text. They were independently generated here, not copied from OEIS b-files.

12.2 The central approximation is excellent for one column, not all

Let $G = e^{F(t_n)+nt_n}/\sqrt{2\pi B}$, with the infinite sums evaluated by high-precision truncation. The following are numerical diagnostics, not interval-certified bounds. Values shown are rounded from the included CSV.

α	n	$a_\alpha(n)/G$	$a_\alpha(n)/(G(1 + \mathcal{E}_1))$	$a_\alpha(n)/(G(1 + \mathcal{E}_1 + \mathcal{E}_2))$
1	100	0.989159702	0.999826713	0.999908591
1	1000	0.997526322	0.999995495	0.999999984
1	5000	0.999084078	0.999999367	0.999999999
2	5000	0.999545155	1.000036354	1.000036527
3	5000	0.961625508	0.961944240	0.961944315
4	5000	1.230518103	1.230825332	1.230825386
5	5000	1.581097698	1.581414178	1.581414223

At $\alpha = 1, n = 5000$, two central corrections reduce the relative error to about 1.1×10^{-9} . At $\alpha = 5, n = 5000$, the central approximation is plainly not accurate, and adding central corrections does not repair it. This is evidence about finite-size behavior, not a contradiction of a fixed-parameter limit. In particular, these tests must not be advertised as numerical confirmation of rapid convergence for all five columns.

[Merge note, batch 75.] Source 75-01's tests at $\alpha = 2, 3, n = 2500$ show the same effect, with residuals that do not change from $J = 1$ to $J = 3$: a noncentral contribution, not slow Edgeworth convergence (Section 16.3).

The resonance diagnostic computes the exact product modulus at $\theta = 2\pi/M$, using the same numerical saddle:

α	n	$ \phi_t(2\pi/M) $
1	5000	2.33482×10^{-15}
2	5000	2.72621×10^{-5}
3	5000	0.0173352
4	5000	0.150645
5	5000	0.361805

The increase in these noncentral moduli is consistent with the observed failure of a central-only approximation. Their values do not by themselves predict the sign of the coefficient error; the omitted phases matter.

12.3 Logarithmic expansions are not monotone numerical improvements

For $\alpha = 1, n = 5000$, the exact logarithm is approximately 401.567349464797. Subtracting the leading expression $K(r - 1)$ leaves 41.05033086599. Subtracting the expansion through

P_3 instead leaves -2.72431695876 , while truncating through P_7 leaves -7.67561845406 . The additional terms are not a promise of monotone improvement at a fixed small r . The missing prefactor and endpoint contributions are also on scales suppressed by the logarithmic theorem but relevant numerically.

An independent continuum check evaluates the integral and mean in Lemma 7.1, without using the rational-function expansion, and compares the normalized remainder after P_7 with P_8 :

α	s	Scaled remainder $r^8(\text{entropy}/(\alpha K) - \mathcal{H}_{\alpha,7}(r))$
1	24	7881.08213
1	48	5997.09948
1	96	5424.47363
Predicted limit $P_8(\pi^2/6)$:		4963.76905
3	24	15.7012553
3	48	9.85270737
3	96	9.49113831
Predicted limit $P_8(\pi^2/54)$:		9.15764483

The approach is slow, but has the expected direction in these samples. These are continuum diagnostics at nonintegral implied n , not exact integer-sequence computations at exponentially large indices.

12.4 What the numerical implementation guarantees

The integer arithmetic and recurrence comparisons are exact. For the saddle calculations, the program uses arbitrary-precision arithmetic and chooses a cutoff from a geometric majorant for $\sum k^{\alpha+6}e^{-tk}$. This bounds the tails relevant to all cumulants through order six, apart from harmless fixed constants. The test used 50 requested decimal digits with guard digits; the saved mean residuals are far smaller than the displayed precision. Nevertheless, the majorant and Newton iteration are evaluated in floating point, not outward-rounded interval arithmetic. Accordingly, the CSV values are diagnostics rather than certificates of rigorous numerical error bounds.

12.5 Source 48's computations for A022629 (source 48)

Two exact coefficient constructions. For an integer exponent $\alpha \geq 1$, multiply the product factors successively. Starting with $a_0 = 1$ and $a_n = 0$ for $n > 0$, process each k by updating in *decreasing* order of n : $a_n \leftarrow a_n + k^\alpha a_{n-k}$ ($n \geq k$). Descending order enforces at most one use of size k ; increasing order would compute a different partition problem. The independent check is the logarithmic-derivative recurrence (12.2). That formal identity is valid regardless of the analytic restriction in Proposition 3.4: only finitely many terms contribute to a fixed degree. The program `code/48-a022629-verify.py` compares the two constructions through degree 100, checks the initial OEIS values and strict monotonicity, and computes the whole exact table through $n = 6400$ for $\alpha = 1$ (`data/48-a022629-exact_coefficients.csv`).

[Merge note, batch 73O1.] On the common range $n \leq 5000$ this table equals source 40's `data/40-norm-moments-A022629_computed.txt` entry for entry; the intake also recomputed both from the product through $n = 6400$.

Saddle diagnostics. The exact saddle is found by Newton iteration, $t_{\text{new}} = t + (-F'(t) - n)/B(t)$, with a positivity safeguard. The first four cumulants use $b_2 = p(1 - p)$, $b_3 = p(1 - p)(1 - 2p)$ and $b_4 = p(1 - p)\{1 - 6p(1 - p)\}$. For $\beta \geq 0$ the explicit tail bound

$$\sum_{k > k_0} k^\beta e^{-tk} \leq e^t t^{-\beta-1} \Gamma(\beta + 1, t(k_0 + 1)) \quad (12.3)$$

follows by integrating $(u + 1)^\beta e^{-tu}$ over $u \geq k_0$. It controls truncation of the log product, mean, variance, and fixed-order cumulants after multiplication by a suitable polynomial constant. This addresses the infinite sums, but the implementation does not turn floating-point rounding into certified interval arithmetic.

n	$\log a(n)$	Gaussian relative error	Corrected relative error
100	31.57034412	$1.096 \cdot 10^{-2}$	$1.733 \cdot 10^{-4}$
400	80.97262500	$4.427 \cdot 10^{-3}$	$1.377 \cdot 10^{-5}$
1600	197.60929471	$1.849 \cdot 10^{-3}$	$2.527 \cdot 10^{-6}$
6400	467.20404426	$7.888 \cdot 10^{-4}$	$4.704 \cdot 10^{-7}$

Table 1: Computed diagnostics for A022629 ($\alpha = 1$), source 48. Relative error means approximation divided by the exact integer coefficient, minus one. The corrected approximation includes the two cumulant terms of (2.9). Calculations used 45 decimal digits.

At $n = 6400$, the bare logarithmic core underestimates $\log a(n)$ by about 45.36. Adding the first four inverse-logarithmic terms instead overestimates it by about 3.55. Thus that four-term logarithmic approximation is still poor as a *relative coefficient* approximation at this size, despite being a valid asymptotic expansion; the corrected exact-saddle formula has relative error approximately $4.7 \cdot 10^{-7}$. This is why Theorems 2.1 and 2.2 are stated separately. The recorded file `data/48-a022629-numerical_checks.json` gives the saddle, variance, cumulant correction, mean residual, cutoff, and tail estimate as well as the errors in Table 1. Its decimal values are computational evidence only; the exact integers and symbolic polynomial checks are exact computations. No Lean executable was run for either source.

[Merge note, batch 7301.] The $n = 100$ row agrees with source 40's $\alpha = 1$ row in Section 12 ($1/0.989159702 - 1 \approx 1.096 \cdot 10^{-2}$, $1/0.999826713 - 1 \approx 1.733 \cdot 10^{-4}$), and source 48's saddle at $n = 100$ agrees with `data/40-norm-moments-run_alpha1.txt` to all printed digits.

13 Further research questions and proposed projects

The proved results leave several substantial and concrete directions.

1. An effective multi-saddle expansion. Locate the secondary saddles near $2\pi\ell/M$, compute their phases and Gaussian factors, and sum the relevant arcs with a uniform remainder. Theorem 11.1 supplies the first exponential action but not the full contribution. The moderate- n failures for $\alpha = 3, 4, 5$ make this a practical as well as theoretical priority.

2. A complete two-scale coefficient transseries. Refine the continuum-to-lattice passage sufficiently to retain endpoint terms, Euler–Maclaurin corrections, the central determinant, and the resonance sectors in one consistent normalization. Determine which of these expansions are Gevrey and whether optimal truncation produces a useful exponential improvement. The present all-logarithmic series alone does not settle this.

3. The crossover as the tilt vanishes. The fixed- α expansion is singular as $\alpha \downarrow 0$, while the unweighted distinct-partition count has a different growth law. A natural crossover parameter is $\alpha \log K$. Derive a uniform family of saddle profiles when this parameter stays bounded, and match it to both the unweighted regime and the fixed-positive-tilt regime. The shrinking-tilt analysis in Gikunda’s thesis provides an important comparison point [10], but matching the full regimes still requires work.

4. Growing moments and concentration near maximal norm. Allow $\alpha = \alpha(n) \rightarrow \infty$. The fluctuation width H and resonance action show that several thresholds can occur before the model collapses onto maximal-norm partitions. Determine the sharp parameter ranges for a local limit theorem, for single-saddle dominance, and for localization near an extremal partition. The fixed- α proof does not justify substituting an arbitrary growing parameter.

5. The joint law of the largest several parts. The independent tail calculation suggests a Poisson process with intensity $e^{-x} dx$ for the rescaled upper parts. Prove its conditioned point-process limit and obtain joint distributions of the largest j parts, including second-order corrections to their centering. The maximum theorem is the one-dimensional starting point, not a substitute for tightness of the process.

6. Fluctuations of the number of parts and of the log norm. A multivariate conditioned local limit theorem should describe the number of occupied sites and the logarithm of the norm under fixed positive tilt. Conditioning removes a covariance direction and can cancel a leading variance term; computing the surviving scale requires more than plugging the unconditioned variance into a Gaussian formula.

7. More general weights. Replace k^α by $k^\alpha (\log k)^\beta$ away from the harmless small indices, or by a smooth regularly varying weight. Identify hypotheses that preserve a single large crossing, the logistic boundary, and a computable inverse expansion. Arithmetic thinning of the allowed part sizes would also require a replacement for the consecutive-block lemma and may introduce persistent lattice obstructions.

8. Certified numerics and formal verification. Give explicit constants in the Fourier bounds, add interval-enclosed saddle solving, and certify a chosen relative-error target at a stated n and α . For Lean, a sensible modular route is: the formal Euler product and recurrence; the Bernoulli coefficient identity; the consecutive-block inequality; an abstract triangular-array local-limit theorem; and finally the boundary smoothing lemma. The algebraic generation of P_j and the inverse coefficients can be separately certified by finite polynomial identities. This would connect directly to the repository’s transseries infrastructure without pretending that formal algebra alone proves analytic asymptotics.

13.1 Source 48’s proposed projects (source 48)

Source 48 lists eight next steps, stated for A022629 and the real family. They are proposed research tasks, not claims that the corresponding problems are all globally open. Its numbers are kept; overlaps with the projects above are marked.

48.1 Resolve all subdominant asymptotic blocks (overlaps project 2). The upper-transition calculation determines every K/r^j term. Determine the smaller contributions

from the lower endpoint and the sum–integral discrepancy, including any powers of K , logarithms, and constants that appear for different α . Prove a fully explicit multiplicative equivalent with no implicit exact saddle sums. Distinguish this task from merely computing more terms in (2.2).

48.2 The transition as $\alpha \downarrow 0$ (overlaps project 3). At $\alpha = 0$ the model is ordinary distinct-part enumeration; for every fixed $\alpha > 0$ the leading logarithm has the different scale $\alpha K \log K$. Develop uniform asymptotics in a double limit, for example when $\alpha \log n$ stays bounded. The present proof explicitly does not cover this regime, and the change in the active interval makes it a useful test case for uniform asymptotic methods.

48.3 The number of parts after conditioning (overlaps project 6). The finite-dimensional occupation profile is simple, but conditioning on total size creates long-range correlations. Determine the mean, variance, and limiting law of the total number of parts to several orders. A naive unconditioned variance calculation is not sufficient because number and size are strongly correlated in the narrow transition region.

48.4 Joint extremes and point processes (overlaps project 5). Upgrade Corollary 10.1 to joint laws of the largest several parts and convergence of the rescaled tail point process. Prove quantitative error bounds, uniform tail estimates, and asymptotics of the expectation and variance of the largest part. Distributional convergence alone does not justify taking expectations.

48.5 Stable weight perturbations (overlaps project 7). Replace k^α by $k^\alpha(\log k)^\beta$ or by $k^\alpha \exp(c_1/k + c_2/k^2 + \dots)$, with a positive convention for small k . Determine which coefficients and extreme-value centring terms are universal and which remember the perturbation. Establish explicit uniform hypotheses rather than assuming regular variation is automatically enough.

48.6 Optimal saddle remainder scales. The absolute cumulant bound gives the safe parameter s/M . Signed cumulant cancellations can make actual errors much smaller, as Table 1 suggests. Determine the sharp first relative correction, its complete expansion, and the optimal error after each correction. This would improve both numerical evaluation and rigorous inverse brackets.

[Batch 75, 2 October 2026.] Partly answered by Section 16 (sources 75-01 and 75-06): the sharp signed scales $\kappa_j \sim \frac{j!}{2} M^{j+1}/(\alpha s)^{j-1}$, the sharp first correction $\mathcal{E}_1 \sim -3/(8\alpha M s)$, and the error $O((\alpha M s)^{-j-1})$ after each correction, with the threshold localization of Theorem 16.4. A complete expansion of $\mathcal{E}_1$ in $1/s$, and optimality beyond these orders, remain open.

48.7 Effective inversion and formalization (overlaps project 8). Turn the asymptotic O constants into explicit bounds, combine them with finite exact verification, and produce a certified algorithm for $N_\alpha(Y)$. A suitable ProveIt formalization can separate finite coefficient algebra, the positive product measure, the block minor-arc estimate, remainder transport, and integer-threshold bracketing. None of those formalization tasks should be replaced by unchecked numerical fits.

[Batch 75, 2 October 2026.] Theorem 16.4 (source 75-01) localizes $N_\alpha(Y)$ to within $1 + o(1)$ of an exact-saddle inverse, but with implicit constants; the explicit bounds asked for here are still open.

48.8 OEIS and literature follow-through. Submit a concise, independently reviewed proof of (1.3) and the first sharper correction to the OEIS editors. Continue the literature

audit for products with coefficients *inside* their factors, rather than mistaking $\prod(1+k^\alpha q^k)$ for $\prod(1+q^k)^{k^\alpha}$. Compare any earlier theorem at the level of hypotheses and error terms before making a priority claim.

14 Conclusion

The five checked OEIS conjectures share a single analytic mechanism. Filling parts up to a moving boundary produces the dominant factorial-scale entropy; smoothing that boundary produces the π^2 correction and its all-orders inverse-logarithmic successors. A conditioned Bernoulli model supplies the local probability that converts this entropy into an actual coefficient, and also gives logistic occupation and a Gumbel extreme. Lambert- W inversion turns the same expansion into a quantitative threshold law.

The resulting package contains explicit proofs rather than numerical extrapolation. It also records an important limitation: central asymptotics can converge extremely slowly for higher moments because noncentral resonances are only suppressed on the scale $\exp[-C_\alpha K/(\log K)^3]$. That proved scale and the visible finite-size discrepancies identify a well-defined next research problem: a phase-sensitive, exponentially improved multi-saddle expansion.

Source 48 reaches the same conclusions for A022629 by partly different routes: an elementary proof with a stronger additive remainder, an explicit differential operator that generates the entire dominant logarithmic block by finite exact algebra, a Lambert-based inverse for the integer threshold, a precise comparison with the largest individual partition weight, and a Gumbel law centred explicitly in $\log \sqrt{2n}$. The remaining distinctions—formal versus convergent series, logarithmic versus relative accuracy, and proved mathematics versus formal-machine verification—are essential parts of the result rather than editorial qualifications (source 48).

[Merge note, batch 75.] Sources 75-01 and 75-06 reach the same conclusions a third and a fourth time and add what follows: the sharp saddle scale and threshold (Section 16), the eventual shape of the sequence (Section 17), and the convolution powers (Section 18).

15 Addendum (batch 75): two further independent sources

Batch 75 of ProveIt's incoming-reports intake delivered two more manuscripts on the family $\prod_{k \geq 1} (1+k^\alpha q^k)$, both dated October 1, 2026 and both pinned to the ProveIt commit 29aca108e. Neither cites sources 40 or 48, and neither cites the other.

- **Source 75-01** (batch-75 manuscript 01), *Norm-weighted distinct partitions: A proof of the A022629 growth conjecture, all-order asymptotics, Lambert inversion, and eventual Jensen hyperbolicity*: the family for real power $s > 0$ (here α), five printed logarithmic coefficients, an exact-saddle expansion with a sharp remainder, two inversions, eventual strict log-concavity, fixed-degree Jensen hyperbolicity, and exact data to $n = 10000$.
- **Source 75-06** (batch-75 manuscript 06), *Norm-weighted distinct partitions: A022629, all-order expansions, and inversion*: the two-parameter family $\prod_{k \geq 1} (1+k^r q^k)^s$, real power $r > 0$ (here α) and integer number of layers $s \geq 1$ (here μ), eight printed logarithmic and seven printed inverse coefficients, the relative saddle expansion, and logistic and Gumbel laws, all for every number of layers.

These are the third and fourth independent proofs in this report of Proposition 3.2 and Theorems 2.1, 2.2, 2.3 (and, for source 75-06, 2.4), with identical coefficients. Following

Section 1.4, none of these results is printed again. Table 2 records where each source proves them; Section 15.2 keeps the parts of their proofs and remarks that differ from the routes already printed. The genuinely new material follows in Sections 16 (sharp saddle scales and the near-integer threshold), 17 (log-concavity and Jensen polynomials) and 18 (convolution powers), and the two sources’ research questions in Section 19. Their notation is translated by the second table of Section 1.5. Source 75-01 is not a version of source 48, although their archive names differ only in letter case (`A022629_research_package.zip` and `A022629_Research_Package.zip`): their texts share 0.87 % of their word 8-grams in each direction, and their theorem sets are different.

15.1 Same results, further independent derivations

[Merge note, batch 75.] The intake checked the agreements of Table 2 in exact rational arithmetic: $b_j(6c) = P_j(c)$ for $j \leq 8$ and $c_j(6c) = P_j(c)$ for $j \leq 5$; and, solving (9.1) afresh with $P_1, \dots, P_8$, the coefficients of E^2 through R_0^{-8} , which equal source 75-06’s $d_j(6c)$ for $j \leq 8$ and source 75-01’s inverse coefficients for $j \leq 5$. Source 75-01’s exact A022629 table (`data/75-01-jensen-exact_s1.json`) equals both earlier tables on their ranges; its tables for $\alpha = 2, 3$ to $n = 2500$ equal source 40’s `data/40-norm-moments-A092484_computed.txt` and `data/40-norm-moments-A265840_computed.txt` there and are not shipped. Source 75-06’s relative errors at $n = 100, 500, 1000, 5000, 10000$ (Table 3) coincide with source 75-01’s to all printed digits, and their $n = 100$ values with those of sources 40 and 48.

Stale statements and priority caveats. Both pins are the ProveIt commit 29aca108e (1 October 2026, 19:03; full identifier in Appendix D). It already contained the archives of sources 40 and 48, unopened, and predates their placement (9df4ba51a). Source 75-01’s sentence “A repository code search for A022629 returned no match” was accurate for that snapshot and is stale now: the repository’s treatment is this report. Source 75-06 makes no such statement, but its repository reading was blind to the same two archives. Likewise, the proof of the five Kotešovec conjectures that both sources present as their principal contribution is, in this report, the third and fourth proof; their new contributions are those of Sections 16–18. Both keep explicit priority caveats, which this report keeps. Source 75-01: its search “located no prior proof of these precise fixed- s statements”, which is “not a claim that no such proof exists anywhere, nor a claim that a longstanding major conjecture in partition theory has been settled”; Kumar and Rana’s 2026 article on statistical inequalities for norm moments [18] “should also be considered in any eventual publication-level priority review”; the Hermite-limit mechanism is Griffin, Ono, Rolén and Zagier’s [17], and no new general Hermite-limit theorem is claimed. Source 75-06: its targeted search “does not establish bibliographic priority”; Lagarias and Sun’s reciprocal-supernorm ensemble [20] is a different ensemble and not a proof of these results; the conditioning method is Fristedt’s [19], applied with every estimate proved for this model. Source 75-06 read the repository’s `Analysis/Transseries` README, and source 75-01 the repository map and the Lambert- W drafts of the Fabius tree [24]; neither uses a repository theorem as a premise, and neither claims Lean or Rocq verification.

15.2 Second routes and remarks that differ

Analytic inverse and Cauchy bounds (source 75-01). Source 75-01’s proof of Lemma 7.1 controls the Taylor remainder of the boundary inverse by analyticity rather than by repeated use of (7.3). The local inverse $u_s(v)$ of h_s is analytic on $|v| \leq cs$ for a fixed small $c > 0$: either by the inverse function theorem with uniform bounds on $u \in [1/2, 3/2]$, or by solving

Result in this report	Source 75-01	Source 75-06	Relation
Proposition 3.2 (two-term growth; the five OEIS conjectures)	Prop. 2.3 with Lemma 2.2	Thm. 2.1 (μ layers)	same lower-bound partition $1, \dots, k-1, k+d$ as sources 40 and 48; third and fourth routes
Lemma 3.1 Strict monotonicity (Section 9) Recurrence (12.2)	Lemma 2.2 Prop. 2.1 (8), $b_s(m) = C_\alpha(m)$ used in Section 4.2	proof of Thm. 2.1 Section 7.1 (10.1), factor μ	same estimate same injection; 75-06 adds $\mu \geq 2$ (Section 18) identical
Lemma 4.1, (4.5)		Lemma 3.1	same; signed version new (Lemma 16.1)
Lemma 5.1, (5.3), (5.4) Theorem 2.2, (6.2), (2.9), (6.3) (5.6) Lemma 7.1, (7.4)–(7.6)	Section 4.1 Thm. 4.1, E_r, E_1, E_2 identical proof of Thm. 1.1 Lemma 5.1, $u_L^{(m)}(0)$ with $a = s-1$	Lemma 5.1, (5.3)–(5.4) Thm. 5.2, Q_j, Q_1 identical Cor. 5.3 Prop. 4.1, $Y_v^{(m)}(0) = U_m(s)$	same geometric-sum proof and bounds same expansion; 75-01's remainder is sharper (Theorem 16.3) same same; 75-01's Taylor remainder by Cauchy bounds (Section 15.2)
Mean relation (8.1), (7.15) (8.2), (8.3), (8.8)	$B_s = Q_\alpha/2$ Section 5.2, $G_s = \mathcal{H}_\alpha$	Prop. 4.2, $1 + d(v) = Q_\alpha$ (6.4)–(6.7)	same; 75-06 proves the mean directly, as source 48 does identical construction
Theorem 2.1, $P_1, \dots, P_8$ (2.5) Theorem 2.3, (9.3), (9.4)	Thm. 1.1, $c_j(6c) = P_j(c)$, $j \leq 5$ (6) Thm. 6.1, through R_0^{-5}	Thm. 6.1, $b_j(6c) = P_j(c)$, $j \leq 8$ (6.3) Thm. 7.1, $\Delta_2, \dots, \Delta_8$	identical; 75-06's b_8 is an exact check of (8.5) same identical through R_0^{-6} ; Δ_7, Δ_8 new (Theorem 18.3)
Theorem 2.4, logistic part Theorem 2.4, Gumbel part, $L_n/K \rightarrow 1 + 1/\alpha$ Remark on the two Lambert branches	— — Section 1.3	Thm. 8.2 Thm. 9.1 Section 7.2	same at $\mu = 1$ same centring equation at $\mu = 1$ same
Signed cumulant scales and $\mathcal{E}_1 \sim -3/(8\alpha Ms)$	Lemma 3.1, Thm. 4.1	Prop. 5.4	each other's duplicate; new here (Section 16)

Table 2: Results of sources 75-01 and 75-06 that this report already proves. Statement numbers are those of the delivered manuscripts; equation numbers in parentheses are the sources' own.

$u - 1 = (v + \log u)/s$ by contraction on a fixed complex disk around 1. Cauchy's estimates then bound the remainder after m Taylor terms of u'_s by a constant times $s^{-m-1}|v|^m$, and truncating the v -integral at cs loses an exponentially small quantity. Derivatives in s are handled by differentiating the localized integral and its explicit remainder before estimating, so no differentiability of an unqualified remainder is assumed.

The mean without differentiating a remainder (source 75-06). Source 75-06 proves the mean relation (8.1) directly, as source 48 does (Lemma 7.2): the odd kernel $(1+e^U)^{-1} - \mathbf{1}_{U<0}$ has moments $2(2j+1)!\eta(2j+2)$, and the identity $\partial_t X = -XX'$ for the upper inverse $X(U)$ of $tx - \alpha \log x$ shows that the directly justified moment series equals the formal negative t -derivative of the boundary expansion. Its first terms reproduce (7.19), including the vanishing coefficient of s^{-4} in $Q_\alpha(s) - 1$ (equivalently of s^{-2} in $\mathcal{B}_\alpha$), which source 75-06 calls “a useful algebraic check”.

Coefficient structure (source 75-06). The formal construction gives $P_j(0) = 0$ and $\deg_c P_j \leq \lfloor (j+1)/2 \rfloor$. Source 75-06 tracks this by assigning degree 2 to c : a term of c -degree a in the normalized entropy cannot begin before $r^{-(2a-1)}$; the initial inverse-derivative series has that filtration, and multiplication, composition and the square-root reversion preserve it. These identities in a formal parameter do not make the analytic theorem uniform as $\alpha \downarrow 0$. $P_1, \dots, P_8$ attain the bound (checked by the intake).

Prefactor bookkeeping (source 75-06). At the saddle, $\log a_\alpha(n) = F(t_n) + nt_n - \frac{1}{2} \log(2\pi B) + O(s/M)$, so the probability prefactor contributes only $O(s)$ to the logarithm, below every scale K/r^R (source 75-06's Corollary 5.3; compare (5.6)).

Two finite-size readings of the logarithmic block. For $\alpha = 1$, $n = 10000$, source 75-01 records $\log a_1(10000) = 613.21132139736055145\dots$. The leading expression $K(r-1)$ is low by about 54.3504; adding the first inverse-logarithmic correction leaves about 7.37119; the five-term truncation is high by about 4.65395. Source 75-06 records the normalized first correction $\mathcal{R}(n) = (r/K)(\log a_1(n) - K(r-1))$, whose limit is $\pi^2/6 \approx 1.6449$ by (2.5): exact values give $\mathcal{R}(100) \approx 1.5450$, $\mathcal{R}(1000) \approx 1.8190$ and $\mathcal{R}(10000) \approx 1.9030$, and saddle evaluations at $n = 10^6, 10^8, 10^{10}$ give about 1.8800, 1.8274 and 1.7900. The last three are saddle estimates, not comparisons with exact coefficients. Both sources draw the conclusion of Section 12: the logarithmic block is not a high-accuracy numerical exponent at these sizes, and plots alone can obscure the correct next coefficient.

16 Sharp saddle scales and near-integer thresholds

This section is source 75-01's Sections 3, 4 and 6.2, translated into the notation of this report, with source 75-06's Proposition 5.4 as a second proof of the cumulant scales. Throughout, t , s , M and H are as in (2.7), and, beside the safe parameter $\varepsilon = s/M$ of (4.8), we use the *sharp* parameter

$$\varepsilon_\# = \frac{1}{\alpha M s} = \frac{\varepsilon}{\alpha s^2}. \quad (16.1)$$

16.1 Signed cumulant scales

Lemma 16.1 (Signed cumulant scales; sources 75-01 and 75-06). *For fixed $\alpha > 0$, as $s \rightarrow \infty$ with $t = \alpha s/M$,*

$$\kappa_1 \sim \frac{M^2}{2}, \quad \kappa_j \sim \frac{j!}{2} \frac{M^{j+1}}{(\alpha s)^{j-1}} \quad (j \geq 2 \text{ fixed}). \quad (16.2)$$

Consequently, at the exact saddle,

$$\frac{\kappa_j}{B^{j/2}} = O_{\alpha,j}(\varepsilon_{\sharp}^{(j-2)/2}), \quad j \geq 3. \quad (16.3)$$

Proof (source 75-01). Sum–integral comparison for $x^j b_j(p(x))$, where $p(x) = (1 + e^{tx - \alpha \log x})^{-1}$, has error $O_{\alpha,j}(M^j)$: differentiate the summand in x , split at $M/2$ and $2M$, and use exponential decay beyond $2M$. For $j \geq 2$, b_j has a factor $p(1-p)$ and its other factors are bounded on $[0, 1]$. For $j = 1$ the bulk integral is $\int_0^M x dx = M^2/2$ up to lower-order errors, which is (4.3).

For $j \geq 2$ the upper moving edge supplies the leading term. On $x \geq M/2$ the substitution $y = tx - \alpha \log x$ is increasing for large s ; let $x(y)$ be its inverse there and $w(y) = e^y/(1 + e^y)^2$. Since $p = (1 + e^y)^{-1}$ and $dp/dy = -p(1-p)$, the recursion $b_{j+1}(p) = p(1-p)b'_j(p)$ gives

$$b_j((1 + e^y)^{-1}) = (-1)^{j-2} w^{(j-2)}(y), \quad j \geq 2.$$

Integration by parts $j - 2$ times transfers all these derivatives to $x(y)^j x'(y)$, and the signs cancel. The boundary terms at $M/2$ are exponentially smaller in s than the edge term, and those at infinity vanish. For bounded y ,

$$x(y) = M\{1 + O_{\alpha,y}(s^{-1})\}, \quad x'(y) = \frac{M}{\alpha s}\{1 + O_{\alpha,y}(s^{-1})\}.$$

Repeated differentiation, using $x' = 1/(t - \alpha/x)$, gives

$$\frac{d^{j-2}}{dy^{j-2}} \{x(y)^j x'(y)\} = \frac{j!}{2} \frac{M^{j+1}}{(\alpha s)^{j-1}} (1 + o_{\alpha,j}(1)) :$$

the leading operator is $t^{-1}d/dx$, and differentiating its denominator costs a relative $O_{\alpha,j}(s^{-1})$. The kernel w has integral one and exponential tails; localization first to $|y| \leq C \log s$, then $C \rightarrow \infty$, justifies integrating the expansion.

The region below $M/2$ contributes $O_{\alpha,j}(M^{j+1-c_\alpha} s^{C_j})$ for some $c_\alpha > 0$ (split again at $x = M^{1/2}$ and use $p(1-p)$ above it), which is smaller than M^{j+1}/s^{j-1} , and so is the sum–integral error. This proves (16.2), and dividing by $B^{j/2}$, with $B = \kappa_2 \sim M^3/(\alpha s)$, gives (16.3). $\square$

Second proof of (16.2) (source 75-06). After the same reduction to the upper branch and the same $j - 2$ integrations by parts, the integrand is $\frac{d^{j-2}}{dy^{j-2}}(x(y)^j x'(y)) w(y)$. On bounded y -intervals $x \sim M$ and $x' \sim A := M/(\alpha s)$, and every occurrence of a higher derivative of x costs a further factor $O(1/s)$. So the leading term comes from the affine function $x = M + Ay$:

$$\frac{d^{j-2}}{dy^{j-2}} ((M + Ay)^j A) \Big|_{y=0} = \frac{j!}{2} M^2 A^{j-1} = \frac{j!}{2} \frac{M^{j+1}}{(\alpha s)^{j-1}}.$$

Exponential decay of w , whose integral is 1, permits restriction to a growing bounded neighbourhood followed by dominated estimates. $\square$

Remark 16.2 (Why the absolute cumulant bound is not sharp; source 75-01). Taking absolute values before the integrations by parts loses the cancellations in b_j and gives M^{j+1}/s , which is (4.5), instead of M^{j+1}/s^{j-1} . The signed estimate is what identifies $\varepsilon_{\sharp} = (\alpha Ms)^{-1}$, rather than $\varepsilon = s/M$, as the algebraic saddle parameter. The weaker absolute estimate is still the right tool for a high-order Taylor remainder, but not for the retained coefficients.

16.2 The sharp exact-saddle remainder

Theorem 16.3 (Sharp exact-saddle expansion; source 75-01). *For fixed $\alpha > 0$ and every fixed integer $J \geq 0$, at the exact saddle $t = t_n$,*

$$a_\alpha(n) = \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}} \left(1 + \sum_{\ell=1}^J \mathcal{E}_\ell + O_{\alpha,J}(\varepsilon_\#^{J+1}) \right), \quad (16.4)$$

with $\mathcal{E}_\ell$ as in (6.2). Here $\mathcal{E}_\ell = O_{\alpha,\ell}(\varepsilon_\#^\ell)$ and

$$\mathcal{E}_1 \sim -\frac{3}{8\alpha Ms} \sim -\frac{3}{8\alpha Kr}. \quad (16.5)$$

Proof (source 75-01). The Fourier reduction and the bounds (5.3) and (5.4) are those of Section 5; outside $|\theta| \leq c_0/M$ the integral is smaller than $B^{-1/2}\varepsilon_\#^A$ for every fixed A . This is an eventual estimate: its unspecified α -dependent constant is not a useful finite- n accuracy certificate.

In the central interval put $x = \theta\sqrt{B}$. The exponent is $-x^2/2 + \sum_{j=3}^{R-1} \kappa_j(ix)^j/(j!B^{j/2})$ plus a Taylor remainder. For each fixed $R \geq 2$ the absolute R th derivative on $|\theta| \leq c_0/M$ is $O_{\alpha,R}(M^{R+1}/s)$: for $k \leq 2M$ the complex Bernoulli denominators stay comparable to their positive values, and for $k > 2M$ the odds are small and exponentially decreasing. Hence the normalized remainder is at most $C_{\alpha,R}(s/M)^{R/2-1}|x|^R$. For the expansion through order J choose $R = 2J + 6$; then $(s/M)^{J+2} = o_\alpha(\varepsilon_\#^{J+1})$, since $s^{2J+3}/M \rightarrow 0$.

Restrict first to $|x| \leq \sqrt{C \log M}$. By (16.3) and the remainder bound, the exponential may be expanded with an integrable Gaussian majorant. Give the j th normalized cumulant weight $j - 2$: a monomial of weight w is $O(\varepsilon_\#^{w/2})$ times a fixed polynomial in x . Retain weights through $2J + 1$. Odd weights give odd polynomials, which integrate to zero on the symmetric interval; the first discarded even weight is $2J + 2$, which contributes $O_{\alpha,J}(\varepsilon_\#^{J+1})$. Expanding sufficiently many ordinary Taylor terms makes this a finite remainder estimate. The rest of $|\theta| \leq c_0/M$ is algebraically negligible by (5.3), for C large. The Gaussian moments $(2\pi)^{-1/2} \int e^{-x^2/2}(ix)^T dx = (-1)^{T/2}(T-1)!!$ for even T give exactly (6.2), and the bound on $\mathcal{E}_\ell$ follows from its weight. Finally, Lemma 16.1 gives

$$\mathcal{E}_1 = \frac{\kappa_4}{8B^2} - \frac{5\kappa_3^2}{24B^3} \sim \frac{12}{8\alpha Ms} - \frac{45}{24\alpha Ms} = -\frac{3}{8\alpha Ms}.$$

□

Theorem 16.3 sharpens Theorem 2.2: after the corrections $\mathcal{E}_1, \dots, \mathcal{E}_{J-1}$ the relative error is $O(\varepsilon_\#^J) = O((\alpha Ms)^{-J})$ rather than $O((s/M)^J)$, a gain of $(\alpha s^2)^J$, and the first correction is negative, of exact order $3/(8\alpha Kr)$, so the bare Gaussian factor overestimates $a_\alpha(n)$. At one layer, source 75-06's Proposition 5.4 states the first refinement

$$a_\alpha(n) = \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}} \left(1 - \frac{3 + o(1)}{8\alpha Ms} \right),$$

deduced from its remainder $O(\varepsilon^2) = o(\varepsilon_\#)$; Section 18 gives its version for μ layers. The second correction $\mathcal{E}_2$ is (6.3) (source 75-01 prints the same display), and source 75-01's program computes $\mathcal{E}_3$ from the general formula (6.2) rather than from a hard-coded expression.

[Batch 75, 2 October 2026.] This theorem and Lemma 16.1 partly answer source 48's project 48.6, "Optimal saddle remainder scales" (Section 13): they give the sharp signed scales, the sharp first relative correction, and the error order $\varepsilon_\#^{J+1}$ after each correction. Neither source gives the complete expansion of $\mathcal{E}_1$ in powers of $1/s$, nor any optimality statement beyond these orders; that part of the question stays open.

16.3 Saddle accuracy at finite size

Let $G_J(n) = G \sum_{\ell=0}^J \mathcal{E}_\ell(t_n)$ with $\mathcal{E}_0 = 1$ and G as in Section 12. Table 3 gives source 75-01's signed relative errors $G_J(n)/a_1(n) - 1$, computed with 40 decimal digits in the finite sums, together with the two extra rows of source 75-06. A 60-digit rerun at $n = 10000$ changes the recorded errors by at most 7.0×10^{-38} ; this stability test is not an interval-arithmetic certificate, and neither source's floating-point computation is directed-rounding certified.

n	$J = 0$	$J = 1$	$J = 2$	$J = 3$
100	$1.09591 \cdot 10^{-2}$	$1.73317 \cdot 10^{-4}$	$9.14176 \cdot 10^{-5}$	$9.01486 \cdot 10^{-5}$
200*	$6.90519 \cdot 10^{-3}$	$2.59790 \cdot 10^{-5}$		
500	$3.84109 \cdot 10^{-3}$	$1.07637 \cdot 10^{-5}$	$1.45599 \cdot 10^{-7}$	$8.06153 \cdot 10^{-8}$
1000	$2.47981 \cdot 10^{-3}$	$4.50477 \cdot 10^{-6}$	$1.57836 \cdot 10^{-8}$	$-2.77601 \cdot 10^{-9}$
2000*	$1.60998 \cdot 10^{-3}$	$1.92292 \cdot 10^{-6}$		
2500	$1.40254 \cdot 10^{-3}$	$1.46485 \cdot 10^{-6}$	$3.61741 \cdot 10^{-9}$	$1.17092 \cdot 10^{-11}$
5000	$9.16761 \cdot 10^{-4}$	$6.32944 \cdot 10^{-7}$	$1.05783 \cdot 10^{-9}$	$2.22571 \cdot 10^{-12}$
10000	$6.02038 \cdot 10^{-4}$	$2.75808 \cdot 10^{-7}$	$3.11999 \cdot 10^{-10}$	$4.66600 \cdot 10^{-13}$

Table 3: Signed relative errors $G_J(n)/a_1(n) - 1$ for A022629 ($\alpha = 1$), source 75-01; rows marked * are source 75-06's ($J \leq 1$ only). At $n = 100, 500, 1000, 5000$ and 10000 source 75-06's $J = 0, 1$ values agree with these to every printed digit.

For larger powers the picture changes. At $\alpha = 2$, $n = 2500$, source 75-01 finds $G_J/a_2 - 1 \approx -2.2561 \cdot 10^{-4}$, $-2.2601 \cdot 10^{-4}$ and $-2.2601 \cdot 10^{-4}$ for $J = 1, 2, 3$; at $\alpha = 3$, $n = 2500$, the errors for $J = 0, 1, 2, 3$ are about 0.05070 , 0.05016 , 0.05016 and 0.05016 . Source 75-01 reads this as slow convergence (“Higher powers can converge slowly”) and states that the residuals “have not been decomposed here into separate oscillatory sectors”.

[Merge note, batch 75.] The residual is independent of J once $\mathcal{E}_1$ is included, so it is not a truncation effect of the central Edgeworth series, which the corrections already resolve to 10^{-9} . It is the noncentral contribution that Theorem 11.1 and Remark 6.1 identify, invisible to every central correction. It agrees in sign and order of magnitude with source 40's rows $\alpha = 2, 3$ at $n = 5000$ in Section 12 (there $G(1 + \mathcal{E}_1)/a_\alpha - 1 \approx -3.6 \cdot 10^{-5}$ and $+0.040$), whose growth with α the resonance moduli of that section track. Neither source computes the phases that would turn this identification into a formula.

16.4 Near-integer localization of the threshold

For real $x > 0$ let $t(x)$ be the unique solution of $-F'(t(x)) = x$, so that $t_n = t(n)$. The inverse function theorem gives the exact rules

$$\frac{dt}{dx} = -\frac{1}{B}, \quad \frac{d\kappa_j(t(x))}{dx} = \frac{\kappa_{j+1}}{B}. \quad (16.6)$$

Define the Gaussian inverse

$$\begin{aligned} \Psi_0(x) &= xt(x) + F(t(x)) - \frac{1}{2} \log(2\pi B(t(x))), \\ \Psi'_0(x) &= t(x) - \frac{\kappa_3}{2B^2} = t(x) + O_\alpha(M^{-2}) \sim \frac{\alpha s}{M} > 0, \end{aligned} \quad (16.7)$$

and let $n_*(Y)$ be the unique sufficiently large solution of $\Psi_0(n_*(Y)) = Y$. As in Theorem 2.3, $N_\alpha(Y)$ is the least n with $\log a_\alpha(n) \geq Y$ (source 75-01 writes $\mathcal{N}_s(A)$ with $A = e^Y$).

Theorem 16.4 (Near-integer localization; source 75-01). *For fixed $\alpha > 0$, as $Y \rightarrow \infty$,*

$$|N_\alpha(Y) - n_*(Y)| \leq 1 + o(1). \quad (16.8)$$

More precisely, at an exact sequence value, with s taken at t_n ,

$$n_*(\log a_\alpha(n)) - n = -\frac{3}{8\alpha^2 s^2}(1 + o_\alpha(1)). \quad (16.9)$$

If Ψ_0 is replaced by $\Psi_1 = \Psi_0 + \mathcal{E}_1(t(x))$, the displacement at $Y = \log a_\alpha(n)$ is $O_\alpha(M^{-1}s^{-3})$.

Proof (source 75-01). Theorem 16.3 with $J = 1$ gives $\log a_\alpha(n) = \Psi_0(n) + \mathcal{E}_1(t_n) + O_\alpha(\varepsilon_\sharp^2)$. Divide this logarithmic error by the slope (16.7) and apply the mean value theorem in an $o(1)$ neighbourhood of n : the leading displacement is

$$\frac{-3/(8\alpha Ms)}{\alpha s/M} = -\frac{3}{8\alpha^2 s^2},$$

which proves (16.9). With Ψ_1 the logarithmic error is $O_\alpha(\varepsilon_\sharp^2)$, giving $O_\alpha(M^{-1}s^{-3})$ after division by the slope. For general Y put $m = N_\alpha(Y)$, so that $\log a_\alpha(m-1) < Y \leq \log a_\alpha(m)$. Monotonicity of the eventual inverse n_* places $n_*(Y)$ between $m-1+o(1)$ and $m+o(1)$, proving (16.8). $\square$

No universal rounding instruction follows from (16.8): a height arbitrarily close to a coefficient threshold can require deciding on which side it lies. An effective finite-input certificate would also need explicit error constants and validated numerical evaluation; source 75-01's inverse routine solves the approximating equation, not that certification problem.

[Merge note, batch 75.] The two steps are instances of the transseries-and-inversion volume [25]: the passage from the logarithmic residual to the displacement is its residual-to-root conversion (`p0:thm:backward-error`) with slope bounds $\Psi'_0 \asymp \alpha s/M$, and the passage to the integer threshold is its staircase theorem (`p0:thm:staircase`, parts (1)–(2)), whose separation condition is exactly the obstruction to a universal rounding rule just described. That separation step is formalized in Lean as `Fabius.staircase_separation` (with `Fabius.staircase_ceil`) [26]; nothing else in this section is formalized. Compared with Section 9, whose brackets carry the fixed-order inverse-logarithmic errors plus an $O(1)$ rounding, the exact-saddle inverse localizes the threshold to $1 + o(1)$, at the price of evaluating the exact saddle. It does not answer project 48.7, which asks for explicit constants.

For computation one can solve directly in t : with $Z(t) = F(t) + t\kappa_1(t) - \frac{1}{2}\log(2\pi B(t))$ and $Z'(t) = -tB + \kappa_3/(2B)$, Newton iteration for $Z(t) = Y$, initialized by (2.10), returns $n_* = \kappa_1(t)$. This uses only positive real sums, not the inversion of a long truncated power series. At $Y = \log a_1(10000)$, source 75-01 finds

$$n_*(Y) = 9999.9838291094228371832520930128038 \dots,$$

a displacement of about -0.0161708905772 , which is $1.02387586805 \dots$ times the predicted leading value $-3/(8s^2)$ at the saddle. This is consistent with (16.9), but a single ratio is not a proof of its limit. (The recorded saddle tables `data/75-01-jensen-saddle_s2_dps40.csv` and `data/75-01-jensen-saddle_s3_dps40.csv` also list this scaled displacement for $\alpha = 2, 3$: about 1.69 and 0.83 at $\alpha = 2$, $n = 1000, 2500$, and about 52 and 118 at $\alpha = 3$, the finite-size residual of Section 16.3 seen through the inverse.)

17 Eventual log-concavity and Jensen hyperbolicity

This section is source 75-01's Section 7. It needs errors much smaller than those of the logarithmic block, and differentiating a pointwise O -term for $\log a_\alpha(n)$ would not be legitimate. Instead a smooth approximant is built from exact cumulants, and the discrete error is controlled separately.

Let

$$D_r(t) = [z^r] \log \left(1 + \sum_{j \geq 1} \mathcal{E}_j(t) z^j \right), \quad \mathcal{L}_J(x) = \Psi_0(x) + \sum_{r=1}^J D_r(t(x)). \quad (17.1)$$

Thus $D_1 = \mathcal{E}_1$, $D_2 = \mathcal{E}_2 - \mathcal{E}_1^2/2$, and every D_r is a finite expression. (Here r is a summation index, not $\log K$.) For each fixed J , Theorem 16.3 gives

$$\log a_\alpha(n) = \mathcal{L}_J(n) + O_{\alpha,J}(\varepsilon_\#^{J+1}). \quad (17.2)$$

Equations (16.6) and (16.2) give, for fixed r and k ,

$$\frac{d^k}{dx^k} \mathcal{E}_r(t(x)), \quad \frac{d^k}{dx^k} D_r(t(x)) = O_{\alpha,r,k}(\varepsilon_\#^r M^{-2k}) : \quad (17.3)$$

differentiate each finite cumulant monomial with $d\kappa_j/dx = \kappa_{j+1}/B$; every differentiation costs $O_{\alpha,j}(M^{-2})$. If $S(x) = xt(x) + F(t(x))$, then $S' = t$ and $S'' = -1/B$, so

$$\mathcal{L}'_J(x) = t + O_\alpha(M^{-2}), \quad \mathcal{L}''_J(x) = -B^{-1} + O_\alpha(M^{-4}), \quad \mathcal{L}_J^{(k)}(x) = O_{\alpha,J,k}(sM^{1-2k}) \quad (k \geq 2), \quad (17.4)$$

locally uniformly when x changes by a bounded amount.

17.1 Strict log-concavity

Theorem 17.1 (Eventual strict log-concavity; source 75-01). *For every fixed $\alpha > 0$,*

$$\log \frac{a_\alpha(n)^2}{a_\alpha(n-1)a_\alpha(n+1)} \sim \frac{1}{B(t_n)} \sim \frac{\alpha \log \sqrt{2n}}{(2n)^{3/2}}. \quad (17.5)$$

In particular $a_\alpha(n)$ is strictly log-concave for all sufficiently large n .

Proof (source 75-01). Take $J = 3$ in (17.2); the error is $O_\alpha(\varepsilon_\#^4) = o_\alpha(s/M^3)$. Taylor's formula and (17.4) give

$$2\mathcal{L}_3(n) - \mathcal{L}_3(n-1) - \mathcal{L}_3(n+1) = B^{-1} + O_\alpha(M^{-4}) + O_\alpha(sM^{-7}).$$

Adding the three discrete errors does not change the leading term, and $B \sim M^3/(\alpha s)$, $M \sim K$, $s \sim r$ give (17.5). $\square$

17.2 Hermite limits of Jensen polynomials

For integers $d \geq 1$ and $n \geq 0$ define the Jensen polynomials

$$\mathcal{J}_\alpha^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} a_\alpha(n+j) X^j, \quad (17.6)$$

and the Hermite polynomials H_d by $e^{Xz-z^2} = \sum_{d \geq 0} H_d(X) z^d/d!$, so that $H_0 = 1$, $H_1 = X$, $H_2 = X^2 - 2$, $H_3 = X^3 - 6X$. (H_d is not the transition width H .)

Theorem 17.2 (Fixed-degree Jensen hyperbolicity; source 75-01). *For every fixed $\alpha > 0$ and fixed integer $d \geq 1$, let $\delta_n = (2B(t_n))^{-1/2}$. Then, coefficientwise in X ,*

$$\frac{\mathcal{J}_\alpha^{d,n}(e^{-t_n}(\delta_n X - 1))}{a_\alpha(n) \delta_n^d} \longrightarrow H_d(X). \quad (17.7)$$

Consequently $\mathcal{J}_\alpha^{d,n}$ has d distinct negative real roots for all sufficiently large n .

Proof (source 75-01). Choose J with $J + 1 > 3d/2$; then the discrete error in (17.2) is $o_\alpha(\delta_n^d)$. Put $\tau_n = \mathcal{L}'_J(n)$ and $\tilde{\delta}_n^2 = -\frac{1}{2}\mathcal{L}''_J(n) > 0$ for large n . Taylor expansion for each $0 \leq j \leq d$ gives

$$\log \frac{a_\alpha(n+j)}{a_\alpha(n)} = \tau_n j - \tilde{\delta}_n^2 j^2 + \sum_{k=3}^d \gamma_k(n) j^k + o_\alpha(\tilde{\delta}_n^d), \quad \gamma_k = \frac{\mathcal{L}_J^{(k)}(n)}{k!},$$

where the next Taylor derivative contributes $o(\delta_n^d)$, and $\gamma_k(n)/\tilde{\delta}_n^k = O_{\alpha,k}((Ms)^{1-k/2}) \rightarrow 0$ for $k \geq 3$. These are the hypotheses of the Hermite-limit criterion of Griffin, Ono, Rolén and Zagier [17, Theorem 3].

The mechanism can also be seen directly. Expand the exponential of the last display as a polynomial in j through degree d . The coefficient of j^k , divided by $\tilde{\delta}_n^k$, tends to the coefficient of z^k in e^{-z^2} . In the normalized Jensen sum the d th finite difference annihilates every polynomial of degree below d , and the surviving coefficients combine to $H_d(X)$; the local $o(\tilde{\delta}_n^d)$ error is exactly what this cancellation needs. Finally (17.4) gives $\tilde{\delta}_n/\delta_n \rightarrow 1$ and $(\tau_n - t_n)/\delta_n \rightarrow 0$, so the affine change of normalized variable tends to the identity and $\tau_n, \tilde{\delta}_n$ may be replaced by t_n, δ_n . Hermite polynomials have distinct real roots; on disjoint small intervals around them the endpoint signs alternate, coefficientwise convergence preserves those signs, and degree d allows no further roots. The Jensen polynomial has positive coefficients and a nonzero constant term, so its real roots are negative. $\square$

Neither the theorem nor its proof gives a bound uniform in d . It does not assert that every Jensen polynomial of every degree is hyperbolic at one fixed index, and it should not be conflated with infinite iterated log-concavity of the sequence. Source 75-01's contribution here is the verification of the Griffin–Ono–Rolén–Zagier hypotheses, including an error scale small enough to survive finite differences; the Hermite-limit mechanism itself is theirs.

17.3 Exact finite data

Source 75-01 evaluates the positive product recursion exactly through $n = 10000$ for $\alpha = 1$ and through $n = 2500$ for $\alpha = 2, 3$, and checks the signed recurrence (12.2) against it through $n = 200$ for each of $\alpha = 1, 2, 3$. For $\alpha = 1$, among the tested indices $1 \leq n \leq 9999$, strict log-concavity fails exactly at

$$n = 1, 2, 4, 6, 8, 11, 13, 16, 17, 19, 23, 26, 29, 32, 34, 38, 40, 41, 43, 46, 53, 59, 64, 76,$$

so every tested index from 77 through 9999 is strictly log-concave. This does *not* prove that 77 is the eventual threshold. For $\alpha = 2$ the tested range $1 \leq n \leq 2499$ has 162 failures, the last at $n = 1033$; for $\alpha = 3$ it has 904 failures, persisting to $n = 2499$, the end of the range (counts from `data/75-01-jensen-exact-audit_s2.json` and `data/75-01-jensen-exact-audit_s3.json`). These observations caution against uniformity in the power.

Exact rational real-root counting gives the numbers of real roots of $\mathcal{J}_1^{d,n}$ in Table 4; every polynomial there was also checked to be squarefree.

n	$d = 2$	$d = 3$	$d = 4$	$d = 5$
100	2	1	0	1
1000	2	3	4	1
5000	2	3	4	5

Table 4: Real roots of the Jensen polynomials $\mathcal{J}_1^{d,n}$ (source 75-01): the onset of hyperbolicity depends on the degree.

[Batch 75, 2 October 2026.] Source 75-06’s research question on shape inequalities (Section 19, item 75-06.9) asks for eventual log-concavity; Theorem 17.1 answers that part for one layer. Ratio monotonicity, higher finite-difference inequalities, and the case of $\mu \geq 2$ layers stay open.

18 Convolution powers

This section is source 75-06’s two-parameter extension (its Sections 1.2, 2 and 4–9 for more than one layer), translated: source 75-06’s power r is α here and its number of layers s is μ . For fixed $\alpha > 0$ and a fixed integer $\mu \geq 1$ put

$$A_\alpha(q)^\mu = \prod_{k \geq 1} (1 + k^\alpha q^k)^\mu = \sum_{n \geq 0} a_{\alpha,\mu}(n) q^n, \quad K_\mu = \sqrt{2n/\mu}, \quad r_\mu = \log K_\mu. \quad (18.1)$$

For $\mu = 1$ this is $a_\alpha(n)$, with $K_1 = K$, $r_1 = r$. An object is an ordered μ -tuple of distinct partitions of total size n , weighted by the product of their α th norm powers; for integral α the weights count colourings. The analysis also applies to real $\alpha > 0$, when the coefficients are positive reals.

OEIS identifications (source 75-06). For $\alpha = 1$ and $\mu = 2, 3$ the coefficients are [A022630](#) and [A022631](#) [21]; for $\alpha = 1$ and fixed $\mu \geq 1$ they are the positive columns of the array [A297321](#) [22]; for integer $\alpha \geq 1$ and $\mu = 1$ they are the positive columns of [A292189](#) [6]. Every statement below is column by column, with the column index fixed; no uniform assertion is made for a growing index.

[Merge note, batch 75.] The intake confirmed on 2 October 2026 that the OEIS names of A022630 and A022631 are $\prod_{m \geq 1} (1 + mq^m)^2$ and $(\cdots)^3$, that their first fifteen terms equal the coefficients computed from the product, and that A297321’s column k is $\prod_{j \geq 1} (1 + jx^j)^k$. Source 75-06’s proposed-notes file further says A297321 cross-references its first 32 columns to A022629–A022660; the intake did not check that.

The μ -layer model. Let $X_{k,i}$ ($k \geq 1$, $1 \leq i \leq \mu$) be independent Bernoulli variables with parameter p_k of (4.1), and $S = \sum_{k,i} kX_{k,i}$. Then $\log \mathbb{E}e^{-\cdot}$ is $\mu F(t)$, the cumulants are $\mu \kappa_j$, and $a_{\alpha,\mu}(n) = e^{\mu F(t) + nt} \mathbb{P}_t(S = n)$; the conditional law given $S = n$ is the weighted law on μ -layer objects of size n . The saddle equation is $-\mu F'(t) = n$.

[Merge note, batch 75.] So the μ -layer saddle at n is the one-layer saddle $t(n/\mu)$ of Section 16.4 at the real argument n/μ , and s , M , H are its one-layer values there. This is why the logarithmic coefficients below do not depend on μ . Source 75-06 does not phrase it this way; its proofs treat μ layers directly.

Theorem 18.1 (Growth and all orders for μ layers; source 75-06). *Fix $\alpha > 0$ and an integer $\mu \geq 1$.*

1. $\log a_{\alpha,\mu}(n) = \mu\alpha K_\mu(r_\mu - 1) + O_{\alpha,\mu}(K_\mu/r_\mu)$.
2. For every fixed $R \geq 1$, with the polynomials $P_j(c)$ of Theorem 2.1,

$$\log a_{\alpha,\mu}(n) = \mu\alpha K_\mu \left(r_\mu - 1 + \sum_{j=1}^R \frac{P_j(c)}{r_\mu^j} + O_{\alpha,\mu,R}(r_\mu^{-R-1}) \right). \quad (18.2)$$

Proof (source 75-06). (1) Lower bound: split n into μ integers $N_1, \dots, N_\mu$ differing by at most one, and use the partition $1, 2, \dots, k-1, k+d$ of Proposition 3.2 in each layer. Each layer contributes $\alpha \log k! = \alpha\sqrt{2N_i}(\log \sqrt{2N_i} - 1) + O_\alpha(\log n)$, and the sum is $\mu\alpha K_\mu(r_\mu - 1) - O_{\alpha,\mu}(r_\mu)$. Upper bound: with $t = \alpha r_\mu / K_\mu$, positivity gives $\log a_{\alpha,\mu}(n) \leq nt + \mu F(t)$. Write $\log(1 + e^u) = u_+ + \log(1 + e^{-|u|})$ with $u = \alpha \log x - tx$: the positive part integrates to $\alpha K_\mu(r_\mu/2 - 1) + O_\alpha(1)$; on $x \geq K_\mu/2$ the slope bound $|u'| \geq \alpha(r_\mu - 2)/K_\mu$ makes the smoothing integral $O_\alpha(K_\mu/r_\mu)$; on $0 < x \leq \sqrt{K_\mu}$ use $\log 2$; on $\sqrt{K_\mu} \leq x \leq K_\mu/2$ convexity gives $u \leq -\alpha r_\mu/2 + O_\alpha(1)$ and a contribution $O(K_\mu^{1-\alpha/2})$. With the sum–integral error $O_\alpha(r_\mu)$ this gives $\mu F(t) = \mu\alpha K_\mu(r_\mu/2 - 1) + O(K_\mu/r_\mu)$, and $nt = \mu\alpha K_\mu r_\mu/2$.

(2) The boundary and mean expansions of Sections 7 and 8 hold for μF and $-\mu F'$ with the factor μ ; at the saddle $K_\mu^2 = M^2 Q_\alpha(s)$, which is (8.2) with r_μ for r , and $\mu F(t) + nt = \mu\alpha K_\mu \mathcal{H}_\alpha(r_\mu)$ to every fixed order. The local probability contributes $O(s)$ to the logarithm (source 75-06's Corollary 5.3), below every scale K_μ/r_μ^R . Taking enough guard terms in the two independently proved expansions gives the remainder. $\square$

Theorem 18.2 (Relative expansion for μ layers; source 75-06). *Fix $\alpha > 0$, an integer $\mu \geq 1$ and $J \geq 0$. At the exact saddle $-\mu F'(t_n) = n$, with $\varepsilon = s/M$,*

$$a_{\alpha,\mu}(n) = \frac{e^{\mu F(t_n) + nt_n}}{\sqrt{2\pi\mu B}} \left(1 + \sum_{\ell=1}^J \mathcal{E}_\ell^{(\mu)} + O_{\alpha,\mu,J}(\varepsilon^{J+1}) \right), \quad (18.3)$$

where $\mathcal{E}_\ell^{(\mu)}$ is (6.2) with every κ_j (including $B = \kappa_2$) replaced by $\mu\kappa_j$, and $\mathcal{E}_\ell^{(\mu)} = O(\varepsilon^\ell)$. Moreover $\mu\kappa_j \sim \mu \frac{j!}{2} M^{j+1} / (\alpha s)^{j-1}$ and

$$\mathcal{E}_1^{(\mu)} \sim -\frac{3}{8\mu\alpha Ms}, \quad a_{\alpha,\mu}(n) = \frac{e^{\mu F(t_n) + nt_n}}{\sqrt{2\pi\mu B}} \left(1 - \frac{3 + o(1)}{8\mu\alpha Ms} \right). \quad (18.4)$$

The proof is that of Theorem 2.2 and Lemma 16.1: a consecutive block of length $\asymp M/s$ around M in any one layer gives (5.3) and (5.4), and the Taylor estimates are summed over the μ copies.

[Merge note, batch 75.] Each monomial of (6.2) has $\sum_j m_j - T/2 = -\ell$, so $\mathcal{E}_\ell^{(\mu)}(t) = \mu^{-\ell} \mathcal{E}_\ell(t)$: the μ -layer corrections are the one-layer ones at $t(n/\mu)$, divided by μ^ℓ . Source 75-06 states the remainder with the safe parameter ε ; the sharp remainder of Theorem 16.3 is proved only for $\mu = 1$.

Monotonicity and exact computation (source 75-06). For $\mu = 1$ and $n \geq 1$, $a_{\alpha,1}(n+1) > a_{\alpha,1}(n)$ by the injection of Section 9; convolution with a positive nondecreasing sequence preserves nondecrease, and the extra summand and the strict inequalities give eventual strict increase for every fixed μ . For integers $\alpha, \mu \geq 1$ the coefficients through degree N follow from the finite product: for each k , repeat μ times the descending update $A[n] \leftarrow A[n] + k^\alpha A[n-k]$

$(n = N, \dots, k)$, which costs $O(\mu N^2)$ integer operations and $O(N)$ stored integers (an arithmetic count, not a bit-complexity bound). Logarithmic differentiation gives

$$n a_{\alpha,\mu}(n) = \mu \sum_{m=1}^n C_{\alpha}(m) a_{\alpha,\mu}(n-m), \quad (18.5)$$

with C_{α} as in (12.1), also as an identity of real formal power series; the alternating divisor sum can be badly conditioned in floating point. Source 75-06's `code/75-06-powers-checks.py` checks the product against (18.5) through $n = 120$ for $\alpha, \mu \in \{1, 2, 3\}$ and the two-layer product against the self-convolution through $n = 200$.

Theorem 18.3 (Threshold inversion for μ layers; source 75-06). *Let $N_{\alpha,\mu}(Y)$ be the least n with $\log a_{\alpha,\mu}(n) \geq Y$, and put*

$$w = W_0\left(\frac{Y}{\mu\alpha e}\right), \quad R_0 = 1 + w, \quad K_0 = \frac{Y}{\mu\alpha w} = e^{R_0}, \quad n_0 = \frac{\mu K_0^2}{2}.$$

For every fixed $R \geq 2$, as $Y \rightarrow \infty$,

$$N_{\alpha,\mu}(Y) = n_0 \left(1 + \sum_{j=2}^R \frac{\Delta_j(c)}{R_0^j} + O_{\alpha,\mu,R}(R_0^{-R-1}) \right), \quad (18.6)$$

with the polynomials of Table 5.

j	$\Delta_j(c)$	source 75-06's $d_j(u)$, $u = 6c$
2	$-2c$	$-u/3$
3	$-2c$	$-u/3$
4	$4c^2 - 2c$	$u(u-3)/9$
5	$\frac{6}{5}c^2 - 2c$	$u(u-10)/30$
6	$-2c - \frac{77}{5}c^2 - 8c^3$	$-u(20u^2 + 231u + 180)/540$
7	$-2c - 57c^2 - \frac{1336}{105}c^3$	$-u(668u^2 + 17955u + 3780)/11340$
8	$-2c - 138c^2 - \frac{6226}{35}c^3 + 16c^4$	$u(140u^3 - 9339u^2 - 43470u - 3780)/11340$

Table 5: Inverse coefficients. $\Delta_2, \dots, \Delta_6$ are those of (9.3) and (9.4); Δ_7 and Δ_8 are new. Source 75-06's $d_j(u)$ are unrelated to $d_{\alpha}(s)$.

Proof (source 75-06). Write $B(r) = \sum_{j \geq 1} P_j(c)r^{-j}$ and $K_{\mu} = K_0 e^h$. The formal inverse equation is $e^h(R_0 + h - 1 + B(R_0 + h)) = R_0 - 1$, that is (9.1) with $E = e^h$, or

$$h = -\log\left(1 + \frac{h + B(R_0 + h)}{R_0 - 1}\right).$$

It has a unique formal solution $h = O(R_0^{-2})$, since the right side has formal derivative $O(1/R_0)$ in h , and $N_{\alpha,\mu}/n_0 = e^{2h}$ gives Table 5. For the integer threshold, use a high finite truncation of Theorem 18.1: its smooth main term has n -derivative asymptotic to $\alpha r_{\mu}/K_{\mu} > 0$; moving n by a large constant times n_0/R_0^{R+1} changes it by order K_0/R_0^R , enough to bracket the retained residual and the forward remainder; monotonicity then brackets the threshold between neighbouring integers, and the $O(1)$ rounding is absorbed. Only explicit smooth approximants are differentiated. $\square$

[**Merge note, batch 75.**] The intake recomputed $\Delta_2, \dots, \Delta_8$ by solving (9.1) with $P_1, \dots, P_8$ in exact rational arithmetic; they equal source 75-06's $d_j(6c)$. As in Section 9, the core $K(\log K - 1) = Y/(\mu\alpha)$ is, after taking logarithms with $w = \log K - 1$, the equation $w + \log w = \log(Y/(\mu\alpha e))$, an instance of the dominant-block theorem `p0:thm:lambert-core` of [25] on the principal branch; the corrections are its perturbed inversion (`p0:thm:perturbed-inversion`) and the integer step its staircase theorem. The principal branch W_0 inverts the growth; the lower branch W_{-1} parametrizes the moving boundary.

Theorem 18.4 (Boundary profile and largest part for μ layers; source 75-06). *Let $\mathbb{P}_n$ be the weighted law on μ -layer objects of total size n , at the saddle of Theorem 18.2.*

1. *Fix distinct reals $x_1, \dots, x_d$ and layers $i_1, \dots, i_d$, and let $k_{n,j} = \lfloor M + Hx_j \rfloor$. The occupation indicators of these site-layer pairs converge jointly to independent Bernoulli variables with parameters $(1 + e^{x_j})^{-1}$; the same holds for any fixed collection of distinct site-layer pairs with convergent rescaled positions, including different layers at one site. A site $\lfloor yM \rfloor$ is occupied with probability tending to 1 for fixed $0 < y < 1$ and to 0 for fixed $y > 1$.*
2. *Let L_n be the largest occupied size in any layer, and b the large solution of*

$$\frac{\mu}{t} b^\alpha e^{-tb} = 1, \quad b = -\frac{\alpha}{t} W_{-1}\left(-\frac{t}{\alpha} \left(\frac{t}{\mu}\right)^{1/\alpha}\right). \quad (18.7)$$

Then $\mathbb{P}_n\{t_n(L_n - b) \leq x\} \rightarrow \exp(-e^{-x})$ for every real x , and $L_n/K_\mu \rightarrow 1 + 1/\alpha$ in probability; for the A022629 ensemble the limit is 2.

At $\mu = 1$ this is Theorem 2.4, and its proof is the proof given there, with the following changes (source 75-06). The local limit estimate of Section 5.3 survives deleting finitely many site-layer variables at sizes $O(M)$ (choose a longest surviving subinterval of the central block, still a fixed fraction of M/s long), or every variable above a cutoff m with $m/M \rightarrow \gamma > 1$ and deleted variance $O(M^2)$; the target may move by $o(\sqrt{\mu B})$. For the maximum, $tb \rightarrow \infty$ and $b/M \rightarrow (\alpha + 1)/\alpha$, the tail intensity is $\mu \sum_{k>m} p_k = (\mu/t)m^\alpha e^{-tm}(1 + O(1/(tm)) + t) + o(1) \rightarrow e^{-x}$ for $m = b + x/t$, and the no-tail probability is $\prod_{k>m} (1 - p_k)^\mu \rightarrow e^{-e^{-x}}$; the deleted tail has mean $O(M)$ and variance $O(M^2)$. The two scales differ for the reason given after Theorem 2.4: a site at yM with $y > 1$ is occupied with probability about $M^{-\alpha(y-1)}$, there are order M/s sites in an extreme window, and the balance occurs at $y = 1 + 1/\alpha$.

Finite-size checks (source 75-06). At $n = 500$ the first-corrected relative errors for $(\alpha, \mu) = (1, 2), (1, 3), (2, 1)$ are about $6.36 \cdot 10^{-6}$, $4.72 \cdot 10^{-6}$ and $1.76 \cdot 10^{-3}$. The less impressive last case is kept deliberately: finite- n behaviour depends on the parameters, and the theorems are not uniform small- n guarantees.

19 Research questions of sources 75-01 and 75-06

Both sources close with research directions, kept here with their own numbers; overlaps with Section 13 are marked, and questions answered in this report are re-scoped. They are proposed tasks, not claims that the problems are globally open.

Source 75-01.

- 75-01.1 Is 77 the exact log-concavity threshold for A022629?** Theorem 17.1 reduces this to effective analytic bounds plus a finite check, but its constants are implicit; an explicit minor-arc bound is the main bottleneck. For the other columns, find the minimal thresholds and their dependence on α .
- 75-01.2 How fast may the Jensen degree grow?** Find a nontrivial $d(n) \rightarrow \infty$ for which hyperbolicity is still guaranteed; this needs simultaneous control of high cumulants, remainders and Hermite root spacing.
- 75-01.3 The crossover as $\alpha \downarrow 0$** (overlaps project 3 and 48.2): connect the central limit regime near zero tilt to the moving edge, for example when $\alpha \log n = O(1)$.
- 75-01.4 Growing moment order** (overlaps project 4): the diagonal A292190 of A292189 is covered by no theorem here; determine where a continuum edge stops being accurate and maximizing partitions take over.
- 75-01.5 The subdominant Fourier sectors** (overlaps project 1): locate the noncentral saddle contributions, their phases and amplitudes, and explain the finite-size residuals at $\alpha = 2, 3$ (Section 16.3). Theorem 11.1, which source 75-01 did not have, supplies the action at each fixed resonance.
- 75-01.6 Extend the weight class** (overlaps project 7 and 48.5), e.g. $k^\alpha(\log(k+1))^\beta$ or regularly varying weights, preserving the cumulant scale, the local expansion and eventual Jensen hyperbolicity.
- 75-01.7 Joint limit laws under norm weighting** (overlaps project 6, 48.3 and 48.4): number of parts and largest part, with the correct conditional fluctuation scales.
- 75-01.8 A validated inverse** (overlaps 48.7 and project 8): combine the tail bound below with interval evaluation and explicit constants in Theorem 16.3, returning the exact threshold or a certified pair of candidates without all earlier coefficients. Source 75-01's tail bound: for $b \geq 0$ and $k_0 + 1 \geq 2b/t$, $\sum_{k > k_0} k^b e^{-tk} \leq (k_0 + 1)^b e^{-t(k_0+1)} / (1 - e^{-t/2})$.
- 75-01.9 Formalize the finite algebra, then the analysis** (overlaps project 8): the coefficient recurrence, the Bernoulli-cumulant recursion, the formula for $\mathcal{E}_r$, the Lagrange coefficients and the formal inverse recurrence first; then the block bound, the sum-integral estimates and the local Fourier remainder. No such formalization is claimed.
- 75-01.10 The signed companion.** A022661, $\prod_{k \geq 1} (1 - kq^k)$, asks whether its coefficient at $n = 9$ is its only zero (Robert Israel, June 2, 2015) [23]. The positivity method does not settle it; competing phases make it a substantially different target.

Source 75-06.

- 75-06.1 Complete the lower-scale transseries** (overlaps project 2 and 48.1): determine the endpoint, discretization and determinant terms jointly; appending only $-\frac{1}{2} \log(2\pi B)$ would be incorrect.
- 75-06.2 Large-order growth and optimal truncation** (overlaps project 2): growth of $P_j(c)$ for fixed c , a Gevrey bound, control by the complex singularities of the boundary inverse, Borel summability and resurgence.
- 75-06.3 Weak norm weighting** (overlaps project 3 and 48.2): $\alpha = \alpha(n) \downarrow 0$ with $\alpha \log n$ bounded or slowly growing.

- 75-06.4 Growing moments and growing layer counts** (overlaps project 4): α or μ depending on n ; the width $M/(\alpha(s-1))$ may stop diverging, and diagonal entries of both arrays are not covered.
- 75-06.5 Slowly varying activities** (overlaps project 7 and 48.5): $k^\alpha \ell(k)$, persistence of the logistic and Gumbel laws, multiscale forward and inverse expansions.
- 75-06.6 Fluctuations in the number of parts** (overlaps project 6 and 48.3): a two-dimensional local theorem for $\prod(1 + zk^\alpha q^k)^\mu$; no conditional variance formula is asserted.
- 75-06.7 The full extreme point process** (overlaps project 5 and 48.4), including multiplicities across layers and joint convergence with the boundary occupations.
- 75-06.8 Quantitative error bounds and certified computation** (overlaps project 8 and 48.7), including when saddle evaluation beats the quadratic exact product.
- 75-06.9 Shape inequalities and discrete regularity**: eventual log-concavity, ratio monotonicity, higher finite-difference inequalities. *[Batch 75, 2 October 2026.]* Eventual strict log-concavity and fixed-degree Jensen hyperbolicity are proved for $\mu = 1$ by source 75-01 (Theorems 17.1 and 17.2); the rest stays open.
- 75-06.10 Large deviations for the norm**: a joint theory for size and log norm, with rate functions under uniform and norm-tilted measures, first uniformly in α on compact intervals.
- 75-06.11 Formal verification in ProveIt** (overlaps project 8): exact identities and rational reversion first; formalizing the elementary proof would give a checked resolution of the leading conjecture before the probability theory.
- 75-06.12 OEIS and bibliographic follow-through** (overlaps 48.8): formula updates with exact quantifiers, and a broader search (norm-weighted strict partitions, multiplicative statistics, variable-activity fermionic products) before any novelty claim; independent review is the next validation step.

A Reproduction guide and file manifest

Besides this source, its compiled PDF and the README, the report directory contains the two sources' programs and recorded outputs under prefixed names.

File	Purpose
<code>code/40-norm-moments-verify.py</code>	Exact integer sequences, independent recurrence, and numerical saddle/Edgeworth diagnostics (source 40).
<code>code/40-norm-moments-derive_series.py</code>	Symbolic generation of the boundary derivatives and logarithmic coefficients.
<code>code/40-norm-moments-derive_inverse.py</code>	Symbolic reversion through six inverse-logarithmic orders.
<code>code/40-norm-moments-check_continuum.py</code>	Independent continuum quadrature diagnostics.
<code>code/40-norm-moments-check_resonances.py</code>	Direct numerical product moduli at the first resonance.
<code>data/40-norm-moments-*</code>	Computed integer lists, CSV diagnostics, symbolic output, execution logs, and the requirements file.
<code>code/48-a022629-coefficients.py</code>	Source 48's operator construction of six forward and six inverse coefficients.
<code>code/48-a022629-verify.py</code>	Source 48's exact table through $n = 6400$ and its saddle diagnostics.
<code>code/48-a022629-Makefile</code>	Source 48's build and verify targets (delivery layout).
<code>data/48-a022629-*</code>	Source 48's exact table, recorded checks, symbolic output, table fragment, and requirements.
<code>48-a022629-PROVENANCE.md</code>	Source 48's sources and verification record.
<code>code/75-01-jensen-verify.py</code>	Source 75-01's exact tables, log-concavity audits and saddle comparisons.
<code>code/75-01-jensen-derive_series.py</code>	Source 75-01's exact check of five forward and five inverse coefficients.
<code>code/75-01-jensen-extra_checks.py</code>	Source 75-01's recurrence, Jensen root counts, precision comparison and Gaussian inverse.
<code>data/75-01-jensen-*</code>	Source 75-01's exact table to $n = 10000$, audits, saddle tables, symbolic output and requirements.
<code>75-01-jensen-BUILD_AND_VALIDATION.txt</code> , <code>75-01-jensen-OEIS_PROPOSED_NOTES.md</code>	Source 75-01's build record and draft OEIS notes.
<code>code/75-06-powers-formal.py</code>	Source 75-06's exact formal algebra: eight forward, seven inverse coefficients.
<code>code/75-06-powers-numeric.py</code> , <code>code/75-06-powers-checks.py</code>	Source 75-06's exact coefficients and saddle estimates; recurrence and convolution checks.
<code>data/75-06-powers-*</code>	Source 75-06's coefficient, numerical, verification and source-audit records and requirements.
<code>75-06-powers-proposed_oeis_updates.txt</code>	Source 75-06's draft OEIS notes.

The mathematical proofs do not require running the programs. The programs write their outputs under the names of the delivered packages, so they should be rerun on a copy of the delivered layout; the README gives the commands. The source uses standard LaTeX packages and no external bibliography processor. No Lean code or machine-checked proof is bundled or implied.

[Merge note, batch 7301.] Source 40's manifest listed its files under their delivered names (`verify.py`, `derive_series.py`, `data/`, `README.md`, `requirements.txt`) with the commands `python verify.py --max-n 5000 --dps 50`, `python derive_series.py`, `python derive_inverse.py`, `python check_continuum.py`, `python check_resonances.py` and two `pdflatex` passes; source 48's were `python code/coefficients.py --order 6` and `python code/verify.py --max-n 6400 --dps 45`. The shipped names are those above; the delivered READMEs are replaced by the report README.

[**Merge note, batch 75.**] Source 75-01’s appendix and source 75-06’s Section 10.4 give these commands, in their delivered layouts:

```
python code/verify.py --exact --s 1 --nmax 10000 # 75-01; also --s 2, --s 3 with --
nmax 2500
python code/verify.py --s 1 --dps 40
python code/verify.py --s 1 --samples 10000 --dps 60
python code/verify.py --s 2 --samples 1000,2500 --dps 40 # and --s 3
python code/derive_series.py
python code/extra_checks.py
python formal.py --order 8 # 75-06
python checks.py
python numerics.py
bash build.sh
```

Their manuscripts, PDFs and READMEs, source 75-01’s `exact_s2.json` and `exact_s3.json`, and source 75-06’s `build.sh` and `SHA256SUMS.txt` are not shipped; the README says how to rerun the programs on a copy.

B Compact proposed OEIS formula updates

For the entry corresponding to $\alpha = 1, \dots, 5$, write $K = \sqrt{2n}$ and $r = \log K$. The following common formula is proved in Theorem 2.1:

$$\log a_\alpha(n) = \alpha K(r - 1) + \frac{\pi^2 K}{6\alpha r} + \frac{\pi^2 K}{6\alpha r^2} + O_\alpha(K/r^3).$$

It is appropriate to link a reviewed proof before changing a database’s conjectural status. This report has not submitted or edited any OEIS entry. For the array [A292189](#), the result holds down every fixed positive column; it makes no uniform assertion about the main diagonal or columns whose index grows with the row.

Source 48’s suggested A022629 addition

Source 48 suggests the following text for independent editorial review; it has not been submitted.

The logarithmic conjecture follows with the stronger estimate $\log a(n) = \sqrt{2n}(\log \sqrt{2n} - 1) + O(\sqrt{n}/\log n)$. More precisely, with $K = \sqrt{2n}$ and $r = \log K$,

$$\log a(n) = K \left(r - 1 + \frac{\pi^2}{6r} + \frac{\pi^2}{6r^2} + \frac{\pi^2/6 - \pi^4/72}{r^3} + O(r^{-4}) \right).$$

A proof, an all-orders coefficient rule, and a separate relative saddle-point formula are supplied in the accompanying research report.

A particularly short proof of the first assertion uses the partition $(1, 2, \dots, m-1, m+d)$ with $n = m(m+1)/2 + d$ for the lower bound, and $a(n) \leq e^{tn} \prod_k (1 + ke^{-tk})$ with $t = \log \sqrt{2n}/\sqrt{2n}$ for the upper bound. Splitting the logarithm into its positive-part contribution and a smoothing remainder of width $O(\sqrt{n}/\log n)$ finishes the argument; see Section 3.1.

[**Merge note, batch 73O1.**] The “accompanying research report” is now this report.

Sources 75-01 and 75-06's suggested notes

Both sources ship draft OEIS notes, not submitted or approved: 75-01-jensen-OEIS_PROPOS ED_NOTES.md and 75-06-powers-proposed_oeis_updates.txt. Beyond the expansion above (source 75-01 through r^{-5} , source 75-06 through r^{-4} , both in the form π^{2j} or u^j), source 75-01 proposes for A022629 the log-concavity gap $\log(a(n)^2/(a(n-1)a(n+1))) \sim \log \sqrt{2n}/(2n)^{3/2}$, eventual hyperbolicity of fixed-degree Jensen polynomials (by the Griffin–Ono–Rolen–Zagier mechanism, which is not new), and the finite observation that $n = 77, \dots, 9999$ are strictly log-concave, explicitly not a proof that 77 is the threshold. Source 75-06 proposes the same formula for the positive columns of A292189 and, with μK_μ for αK and $u = \pi^2$, for the positive columns of A297321. Both insist that a fixed truncation of a logarithmic expansion is not a relative asymptotic, and both ask for review and a broader priority search before use.

C Verification boundaries and source audit of source 48

Item	Boundary of the claim
OEIS status	A022629 was retrieved with the 2018 formula still labeled a conjecture. This is evidence about that entry, not a guarantee of absence of a proof elsewhere.
Repository inspection	The repository README and the Transseries and Inversion README were read; a targeted A022629 code search returned no match. No complete audit of every research artifact is claimed.
Mathematical results	Conventional proofs are given in the body of this article. The proof of the displayed OEIS conjecture is independent of the more elaborate local limit and formal-expansion arguments.
Symbolic calculations	Six forward and six inverse coefficients are generated by exact rational-polynomial arithmetic in c using the stated operator recurrence.
Integer computations	The product recurrence is evaluated through $n = 6400$; the independent logarithmic-derivative recurrence is checked through $n = 100$.
Floating-point diagnostics	The saddle comparisons use 45 decimal digits and explicit analytic truncation bounds. They are not interval-certified evaluations.
Formal verification	No Lean or Rocq/Coq proof of the new analytic theorems is included or claimed. Repository formalization material is a possible integration path, not a certificate of this article.
Novelty	No worldwide priority or “breakthrough” status is certified. The contribution is the supplied proof and its stated extensions.

D Provenance and merge record

Source	Archive (arrival commit)	Main file, PDF	Pin	Author line
40 (base)	OEIS_Distinct_Partition_Norms.zip (f8c3a392a)	article.tex, 22 pp.	1f1981f68	Research report prepared for Vladimir Reshetnikov
48	A022629_Research_Package.zip (3a9518c52)	article.tex, 20 pp.	dc1a7242d	Research report prepared for Vladimir Reshetnikov; developed with AI assistance; proofs and computations supplied for review
75-01	A022629_research_package.zip (4b874cea0)	article.tex, 20 pp.	29aca108e	Research report prepared for Vladimir Reshetnikov (no AI wording)
75-06	OEIS_Norm_Weighted_Partitions.zip (4b874cea0)	article.tex, 26 pp.	29aca108e	Prepared for Vladimir Reshetnikov (no AI wording)

Both are manuscripts of batch 73 (cluster O1) of ProveIt’s incoming-reports intake, numbered 40 and 48 there; the placement commit `9df4ba51a` staged their files and deleted the archives, which survive in their arrival commits. Source 40 arrived first. Neither cites the other, and their texts share almost nothing (8-word shingle containment 0.004 both ways; the longest common run is a `pdflatex` command line): they are independent. Both pins are ProveIt commits of 1 October 2026; source 48 calls its identifier a “tree response” and does not assert that it pins every subsequent read.

What each contributed. Source 40: the base text, in its order, and everything in it, including the five sequences, P_7 , P_8 , the resonance theorem and the finite-size analysis. Source 48: the second proofs of Proposition 3.2, Lemmas 4.1 and 5.1, Theorem 2.2 and the threshold transfer; Propositions 3.4 and 3.5; Lemma 3.6; the operator route; Corollaries 8.1 and 10.1; the A022629 forms; its computations, projects, OEIS wording and audit table.

Where the merge chose.

- Base: source 40, the more general manuscript (five sequences, one more coefficient, the resonance theorem).
- Notation: source 40’s throughout; source 48’s symbols renamed as in Section 1.5.
- Source 48’s λ -centred Gumbel law is printed as a corollary of Theorem 2.4 with its own proof, since its centring is explicit where source 40’s is the exact root $b(t_n)$.
- Source 48’s statements that duplicate source 40’s (its Theorem 1.1, Theorem 2.4, Lemma 3.1, Lemma 3.2, Theorem 3.3, Theorem 6.1, Proposition 7.2) are printed once, credited, with only their different proofs or forms kept. Its abstract, title page and package-contents paragraph are replaced by this text and the README.
- Both data tables of A022629 are shipped (they agree on $n \leq 5000$).
- The two bibliographies are merged; source 48’s OEIS item is source 40’s [1].

Batch 75. Sources 75-01 and 75-06 are manuscripts 01 and 06 of batch 75 of the intake, both in the arrival commit `4b874cea0`; the placement commit `6ea60e367` staged their files under the prefixes `75-01-jensen-` and `75-06-powers-` and deleted the archives, which survive in the arrival commit. The prefixes continue this report's source-numbered scheme, with the batch named so that `01-` and `06-` cannot be read as batch-73 manuscript numbers. Both pin `29aca108ed25d714f31b1316512777fbbdc8e006` (1 October 2026), which already held the archives of sources 40 and 48 unopened. Their texts share 0.55–1.41 % of their word 8-grams with sources 40 and 48 and with each other (preamble, OEIS references and standard displays): they are independent.

What each contributed. Source 75-01: Lemma 16.1 and Remark 16.2, Theorems 16.3, 16.4, 17.1 and 17.2, the smooth approximant of Section 17, the finite data of Tables 3 and 4, the analytic-inverse route of Section 15.2, ten research questions and its OEIS notes. Source 75-06: the second proof of Lemma 16.1, Section 18 (the μ -layer theorems, Δ_7 , Δ_8 , the OEIS identifications), the exact b_8 , the coefficient-structure remark, the finite-size readings, twelve research questions and its OEIS notes.

Where the batch-75 merge chose.

- Placement: both sources were added to this report rather than filed separately; a separate report would have repeated Sections 2–10. The convolution powers were not split off either: their coefficients are the same P_j , and every proof is this report's at n/μ .
- Duplicates: every result already proved here is recorded in Table 2 and not reprinted; second-route text is kept only where the argument differs (Section 15.2, the second proof of Lemma 16.1, and the μ -layer modifications of Section 18).
- Placement in the text: the new sections follow the conclusion, so that every earlier section, theorem, equation and table keeps its number; unnumbered merge notes in the earlier sections point forward.
- Base for the shared new results: source 75-01 for the sharp cumulant scales and the remainder theorem (it proves the full remainder), source 75-06's Proposition 5.4 as the second proof.
- Notation: this report's throughout; both sources' symbols are renamed as in the second table of Section 1.5. New symbols: μ , K_μ , r_μ , $\varepsilon_\sharp$, n_* , Δ_j , $\mathcal{J}$, H_d , τ_n , γ_k , $\mathcal{R}(n)$.
- Source 75-01's finite-size residuals at $\alpha = 2, 3$, which it reads as slow convergence, are kept with its own words and identified by a merge note as the noncentral contribution of Section 11.
- Both sources' novelty paragraphs and abstracts are replaced by Section 15; their priority caveats are kept there.
- Source 75-01's tables for $\alpha = 2, 3$ duplicate source 40's and are not shipped; its $\alpha = 1$ table, which extends the range to $n = 10000$, is.

References

- [1] OEIS Foundation Inc., *A022629: Expansion of $\prod_{m \geq 1} (1 + mq^m)$* . Conjectural formula attributed to V. Kotešovec, May 8, 2018. Internal entry consulted October 1, 2026; sequence revision #54, March 13, 2026. <https://oeis.org/A022629/internal>.
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- [3] OEIS Foundation Inc., *A265840: Expansion of $\prod_{k \geq 1} (1 + k^3 q^k)$* . Conjectural formula attributed to V. Kotešovec, December 27, 2020. Consulted October 1, 2026. <https://oeis.org/A265840>.
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- [5] OEIS Foundation Inc., *A265842: Expansion of $\prod_{k \geq 1} (1 + k^5 q^k)$* . Conjectural formula attributed to V. Kotešovec, December 27, 2020. Consulted October 1, 2026. <https://oeis.org/A265842>.
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- [11] V. Reshetnikov, *ProveIt*, repository snapshot `1f1981f682b2878bde51a6ad40c22777f362fc05`, *Analysis/Transseries/README.md*, consulted October 1, 2026. <https://github.com/VladimirReshetnikov/ProveIt>.
- [12] V. Reshetnikov, *ProveIt*, repository and the documentation at *Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/README.md*. <https://github.com/VladimirReshetnikov/ProveIt>. Repository tree response inspected during preparation of source 48: `dc1a7242d2ab4a4496b7dfee63256ee767b21fc9` (a commit); the documentation was read from the default branch, so that identifier is not asserted to pin every subsequent read. Consulted October 1, 2026.
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- [23] OEIS Foundation Inc., *A022661: Expansion of $\prod_{m \geq 1} (1 - mq^m)$* , including the zero question attributed to Robert Israel, June 2, 2015. <https://oeis.org/A022661>. Inspected by source 75-01, October 1, 2026.
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- [25] [Added in the batch-75 merge.] ProveIt, *Transseries and Inversion* volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, Theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion`, `p0:thm:staircase` and `p0:thm:backward-error`.
- [26] [Added in the batch-75 merge.] ProveIt, Lean file `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`, declarations `Fabius.staircase_ceil` and `Fabius.staircase_separation`.